Chapter 114

Particle Nodes

This chapter details the Particle nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

Contents

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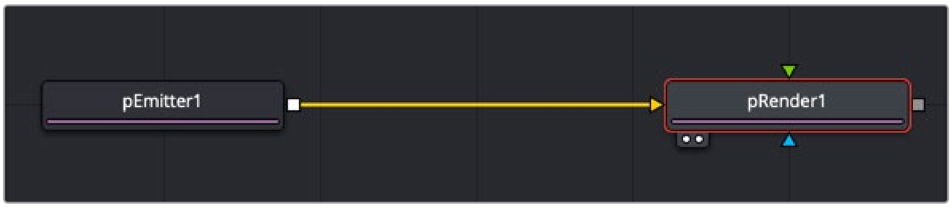
Particle Nodes

The Particle nodes are used to generate a large number of duplicated objects that automatically animate. They are used to create elements like falling rain, fireworks, smoke, pixie dust, and much more. There are endless possibilities. Particles in Fusion consist of a set of nodes that are strung together in a chain for generating, modifying, and rendering particles in a 2D or 3D scene.

To begin, every particle system you create must contain two fundamental nodes:

- **pEmitter:** Used to generate the particles and control their basic look, motion. and behavior.
- **pRender:** Used to render the output of the pEmitter into a 2D or 3D scene. When creating particles, you only ever view the pRender node.



Particle Nodes

The remaining particle nodes modify the pEmitter results to simulate natural phenomena like gravity, flocking, and bounce. The names of particle nodes all begin with a lowercase p to differentiate them from non-particle nodes. They can be found in the particles category in the Effects Library.

pAvoid [pAv]

The pAvoid node

pAvoid Node Introduction

The pAvoid node is used to create a region or area within the image that affected particles attempt to avoid entering and crossing.

It has two primary controls. The first determines the distance from the region a particle should be before it begins to move away from the region. The second determines how strongly the particle moves away from the region.

A pAvoid node creates a “desire” in a particle to move away from a specific region. If the velocity of the particle is stronger than the combined distance and strength of the pAvoid region, the particle’s desire to avoid the region does not overcome its momentum and the particle crosses that region regardless.

Inputs

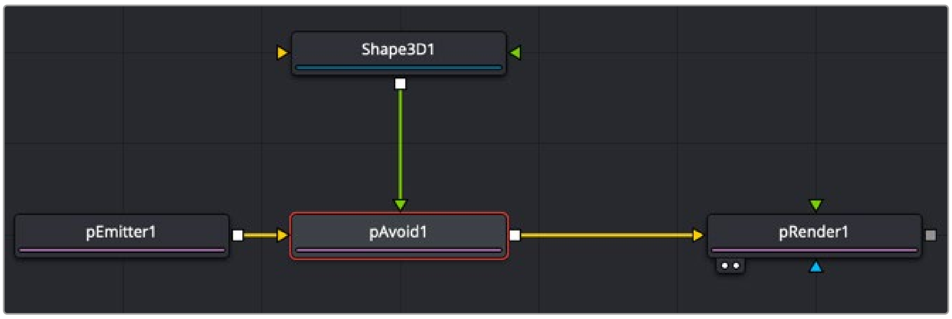
The pAvoid node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area particles avoid.

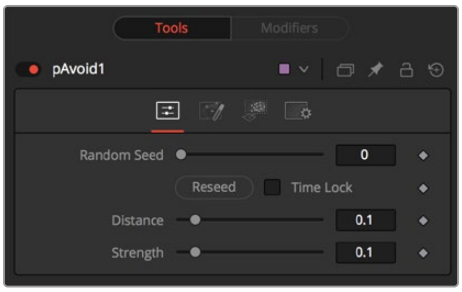
Basic Node Setup

The pAvoid node is placed in between the pEmitter and pRender. A Shape 3D node is used to create the region the particles will avoid.



A pAvoid node using a Shape 3D node as the region to avoid

Inspector



The pAvoid controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Distance

Determines the distance from the region a particle should be before it begins to move away from the region.

Strength

Determines how strongly the particle moves away from the region. Negative values make the particles move toward the region instead.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pBounce [pBn]

The pBounce node

pBounce Node Introduction

The pBounce tool is used to create a region from which affected particles will bounce off when they come into contact.

Inputs

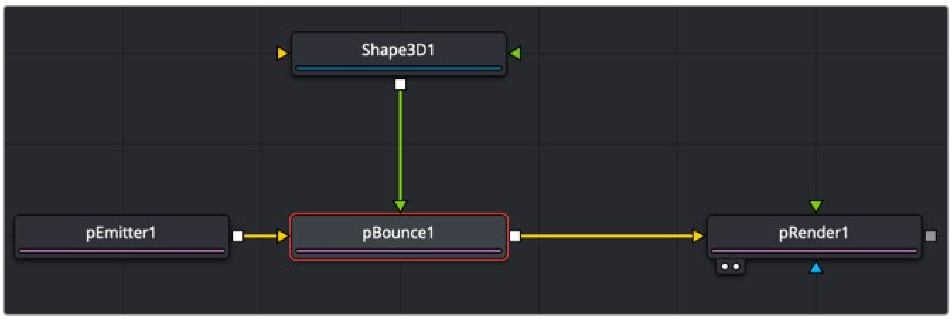
The pBounce node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green or magenta bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area particles bounce off.

Basic Node Setup

The pBounce node is placed in between the pEmitter and pRender. A Shape 3D node is used to create the region the particles bounce off.



A pBounce node using a Shape 3D node as the region on which particles bounce off

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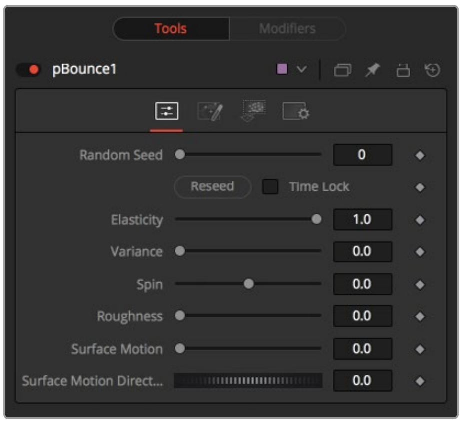
Inspector



The pBounce controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result.

Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Elasticity

Elasticity affects the strength of a bounce, or how much velocity the particle will have remaining after impacting upon the Bounce region. A value of 1.0 will cause the particle to possess the same velocity after the bounce as it had entering the bounce. A value of 0.1 will cause the particle to lose 90% of its velocity upon bouncing off of the region.

The range of this control is 0.0 to 1.0 by default, but greater values can be entered manually. This will cause the particles to gain momentum after an impact, rather than lose it. Negative values will be accepted but do not produce a useful result.

Variance

By default, particles that strike the Bounce region will reflect evenly off the edge of the Bounce region, according to the vector or angle of the region. Increasing the Variance above 0.0 will introduce a degree of variation to that angle of reflection. This can be used to simulate the effect of a rougher surface.

Spin

By default, particles that strike the region will not have their angle or orientation affected in any way. Increasing or decreasing the Spin value will cause the Bounce region to impart a spin to the particle based on the angle of collision, or to modify any existing spin on the particle. Positive values will impart a forward spin, and negative values impart a backward spin. The larger the value, the faster the spin applied to the particle will be.

Roughness

This slider varies the bounce off the surface to slightly randomize particle direction.

Surface Motion

This slider makes the bounce surface behave as if it had motion, thus affecting the particles.

Surface Motion Direction

This thumbwheel control sets the angle relative to the bounce surface.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pChangeStyle [pCS]

The pChange Style node

pChange Style Node Introduction

The pChange Style node provides a mechanism for changing the appearance or style of particles that interact with a defined region. The primary controls mirror those found in the Style tab of the pEmitter node. Particles that intersect or enter the defined region change based on the parameters of this node.

Except for the pCustom node, this is the only node that modifies the particles’ appearance rather than its motion. It is often used to trigger a change in the appearance in response to some event, such as striking a barrier.

Inputs

The pChange Style node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green or magenta bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the custom particle node takes effect.

Basic Node Setup

Opposite of what you may think, to create a change in style that appears to be caused by some physical event, the pChange Style node should be placed before the node that creates the event. For example, below, the particles generated by the Emitter node change style after bouncing off a pBounce. Both the pChange Style and pBounce use the same Shape 3D node as the region. The pChange Style must be placed before the pBounce. If the pChange Style node is placed after the pBounce, the particles bounce off the region before the pChange Style calculates its effect. The particle will never get to intersect with the pChange Style node’s region, and so the style never changes.

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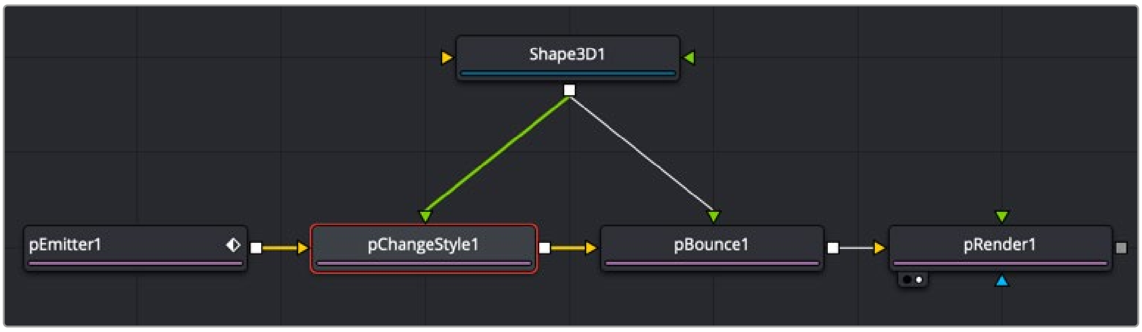
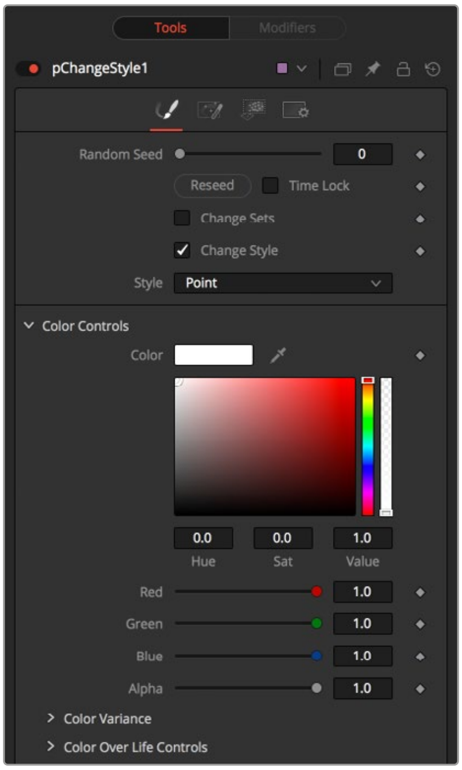


A pChange Style node placed before the pBounce node

Inspector



The pChange Style controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Change Sets

This option allows the user to change the particle's Set to become influenced by forces other than the original particle. See "The Common Controls" in this chapter to learn more about Sets.

Style

This option allows the user to change the particle's Style and thus the look. See "The Common Controls" in this chapter to learn more about Styles.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pCustom [pCu]

The pCustom node

pCustom Node Introduction

The pCustom node is used to create custom expressions that affect the properties of particles. This node is similar to the Custom node, except the calculations affect particles rather than pixels.

Inputs

The pCustom node has three inputs. Like most particle nodes, this orange input accepts only other particle nodes. The green and magenta inputs are 2D image inputs for custom image calculations. Optionally, there are teal or white bitmap or mesh inputs, which appear on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

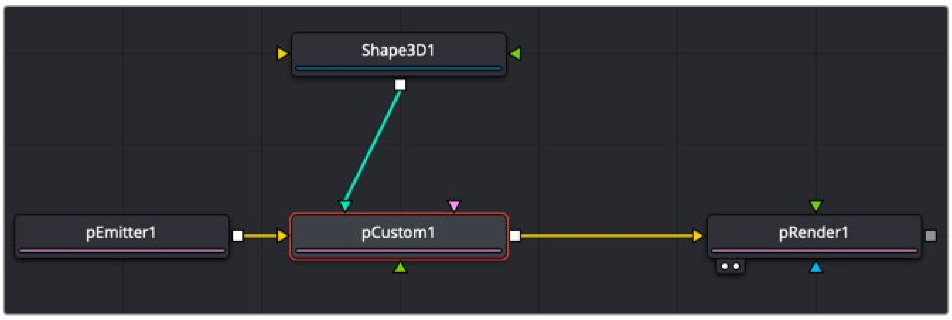
Input: The orange input takes the output of other particle nodes.

Image 1 and 2: The green and magenta image inputs accept 2D images that are used for per pixel calculations and compositing functions.

Region: The teal or white region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the custom particle node takes effect.

Basic Node Setup

The pCustom node is placed in between the pEmitter and pRender. A Shape 3D node is used to create the region where the Custom particle event occurs.

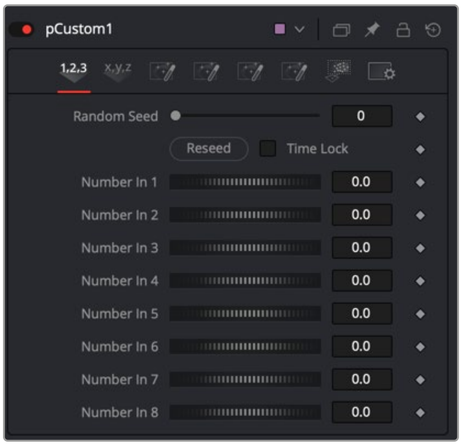


A pCustom node using a Shape 3D node as the region where the custom event occurs

FUSION

Inspector

All the same operators, functions, and conditional statements described for the Custom node apply to the pCustom node as well, including Pixel-read functions for the two image inputs (e.g., getr1w(x,y), getz2b(x,y), and so on).



The pCustom controls

Number 1-8

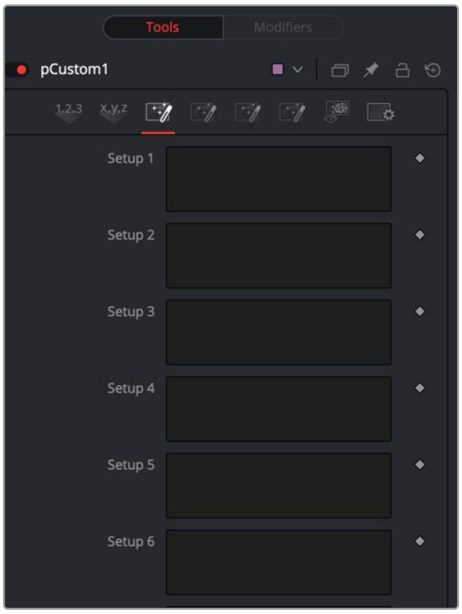
Numbers are variables with a dial control that can be animated or connected to modifiers exactly as any other control might. The numbers can be used in equations on particles at current time: n1, n2, n3, n4, … or at any time: n1_at(float t), n2_at(float t), n3_at(float t), n4_at(float t), where t is the time you want. The values of these controls are available to expressions in the Setup and Intermediate tabs.

pCustom Position tab

FUSION

Position 1-8

These eight point controls include 3D X,Y,Z position controls. They are normal positional controls and can be animated or connected to modifiers as any other node might. They are available to expressions entered in the Setup, Intermediate, and Channels tabs.

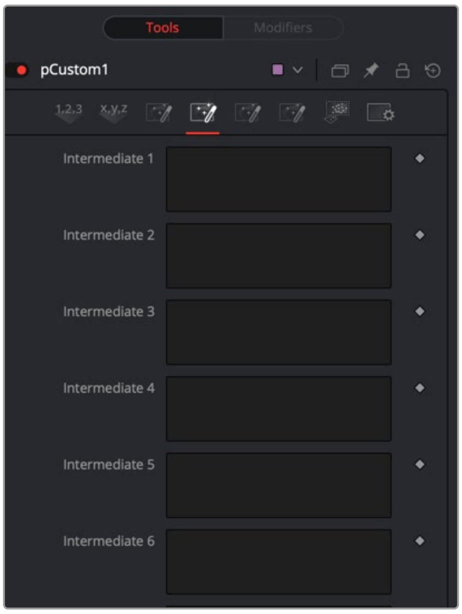


pCustom Setup tab

Setup 1-8

Up to eight separate expressions can be calculated in the Setup tab of the pCustom node. The Setup expressions are evaluated once per frame, before any other calculations are performed. The results are then made available to the other expressions in the node as variables s1, s2, s3, and s4.

Think of them as global setup scripts that can be referenced by the intermediate and channel scripts.



pCustom Intermediate tab

Inter 1-8

An additional eight expressions can be calculated in the Intermediate tab. The Intermediate expressions are evaluated once per frame, after the Setup expressions are evaluated. Results are available as variables i1, i2, i3, i4, i5, i6, i7, i8, which can be referenced by channel scripts.

Particle

Particle position, velocity, rotation, and other controls are available in the Particle tab.

The following particle properties are exposed to the pCustom control:

px, py, pz	particle position on the x, y, and z axis
vx, vy, vz	particle velocity on the x, y and z axis
rx, ry, rz	particle rotation on the x, y, and z axis
sx, sy, sz	particle spin on the x, y, and z axis
pxi1, pyi1	the 2d position of a particle, corrected for image 1's aspect
pxi2, pyi2	the 2d position of a particle, corrected for image 2's aspect
mass	not currently used by anything
size	the current size of a particle
id	the particle's identifier
r, g, b, a	the particles red, green, blue and alpha color values
rgnhit	this value is 1 if the particle hit the pCustom node's defined region
rgndist	this variable contains the particles distance from the region
condscale	the strength of the region at the particle's position
rgnix, rgniy, rgniz	values representing where on the region the particle hit
rgnnx, rgnny, rgnnz	region surface normal of the particle when it hit the region
w1, h1	image 1 width and height
w2 h2	image 2 width and height
i1, i2, i3, i4	the result of the intermediate calculations 1 through 4
s1, s2, s3, s4	the result of the setup calculations 1 through 4
n1..n8	the values of numeric inputs 1 through 8
p1x, p1y, p1z .. p4x, p4y, p4z	the values of position inputs 1 through 4
time	the current time or frame of the compositions
age	the current age of the particle
lifespan	the lifespan of the current particle

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

FUSION

pCustomForce [pCF]

The pCustom Force node

pCustom Force Node Introduction

The pCustom Force node allows you to change the forces applied to a particle system or subset. This node is one of the most complex and the most powerful node in Fusion. If you are experienced with scripting or C++ programming, you should find the structure and terminology used by the Custom Force node to be familiar.

The forces on a particle within a system can have their positions and rotations affected by forces. The position in XYZ and the Torque, which is the spin of the particle, are controlled by independent custom equations. The Custom Force node is used to create custom expressions and filters to modify the behavior. In addition to providing three image inputs, this node will allow for the connection of up to eight numeric inputs and as many as four XY position values from other controls and parameters in the node tree.

Inputs

The pCustom Force node has three inputs. Like most particle nodes, this orange input accepts only other particle nodes. A green and magenta are 2D image inputs for custom image calculations. Optionally there are teal or white bitmap or mesh inputs, which appear on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

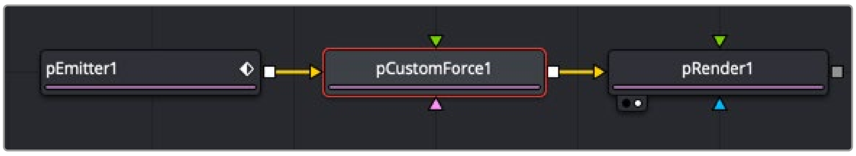
Input: The orange input takes the output of other particle nodes.

Image 1 and 2: The green and magenta image inputs accept 2D images that are used for per-pixel calculations and compositing functions.

Region: The teal or white region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the pCustom Force takes effect.

Basic Node Setup

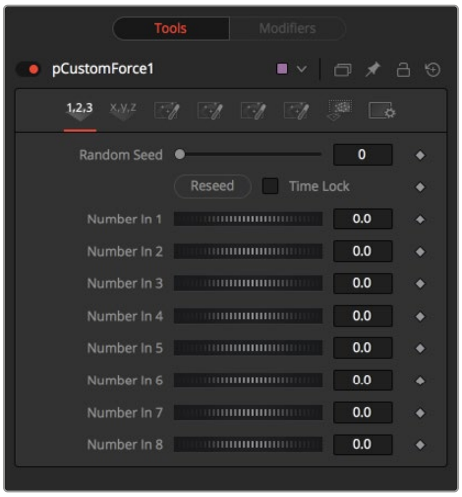
The pCustom Force node is inserted between a pEmitter and pRender node to serve as a catalyst for particles using advanced C++ and scripting.



A pCustom Force is applied to the particles generated by the pEmitter.

FUSION

Inspector



The pCustom Force controls

The tabs and controls located in the Inspector are similar to the controls found in the pCustom node. Refer to the pCustom node in this chapter for more information.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pDirectionalForce [pDF]

The pDirectional Force node

pDirectional Force Node Introduction

This node applies a unidirectional force that pulls the affected particles in a specified direction. Its primary controls affect the strength of the force, and the angle of the force’s pull along the X, Y, and Z axis.

Since the most common use of this node is to simulate gravity, the default direction of the pull is down along the Y axis (-90 degrees), and the default behavior is to ignore regions and affect all particles.

Inputs

The pDirectional Force node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green or magenta bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

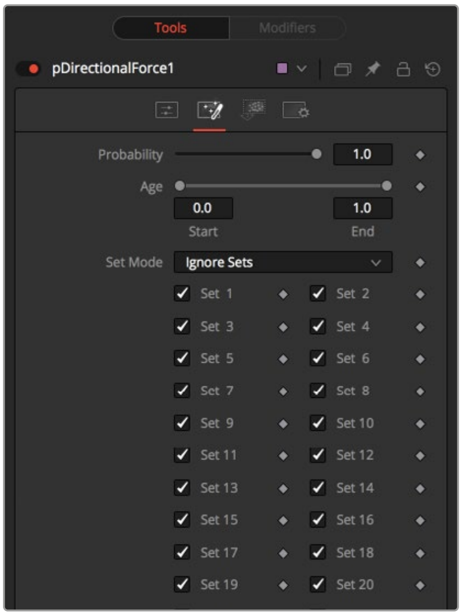
Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the directional force takes effect.

Basic Node Setup

The pDirectional Force node is placed in between the pEmitter and pRender and is often used to create gravity.

A pDirectional Force node placed between the pEmitter and pRender nodes

Inspector



The pDirectional Force controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to select a new seed value randomly, or adjust the slider to select a new seed value manually.

Strength

Determines the power of the force. Positive values will move the particles in the direction set by the controls; negative values will move the particles in the opposite direction.

Direction

Determines the direction in X/Y space.

Direction Z

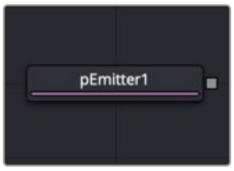
Determines the direction in Z space.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pEmitter [pEm]



The pEmitter node

pEmitter Node Introduction

The pEmitter node is the main source of particles (pImage Emitter is another) and will usually be the first node used in any new particle system. This node contains controls for setting the initial position, orientation, and motion of the particles, as well as controls for the visual style of each particle.

Like all other Particle nodes (with the exception of the pRender node), the pEmitter produces a particle set, not a visible image, and therefore cannot be displayed directly in a viewer. To view the output of a particle system, add a pRender node after the pEmitter.

Inputs

By default, the pEmitter node has no inputs at all. You can enable an image input by selecting Bitmap from the Style menu in the Style tab. Also, two region inputs, one for bitmap and one for mesh, appear on the node when you set the Region menu in the Region tab to either Bitmap or Mesh. The colors of these inputs change depending on the order in which they are enabled.

Style Bitmap Input: This image input accepts a 2D image to use as the particles’ image. Since this image duplicates into potentially thousands of particles, it is best to keep these images small and square—for instance, 256 x 256 pixels.

Region: The region inputs takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the particles are emitted.

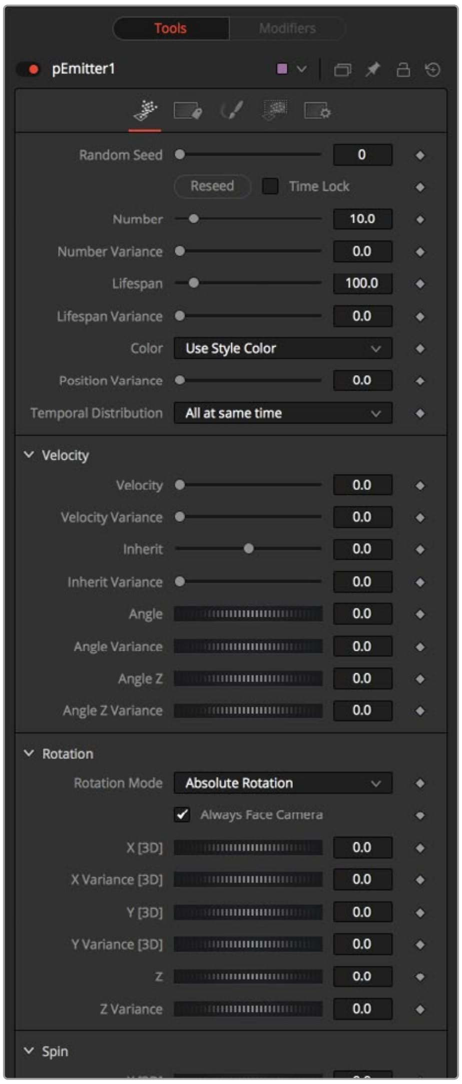
Basic Node Setup

The pEmitter node starts the branch of a particle system that always ends with a pRender node. The pEmitter can feed directly into a pRender node to feed other particle nodes.

A pEmitter node connected to a pRender node is a typical setup for more particle systems.

Inspector

The pEmitter inspector is divided into four main tabs and a common settings tab. The controls tab is the first tab displayed and it contains settings that affect the general setup of the particle cells emitted by the node. These settings do not directly affect the appearance of the cells or shape of the emitter region. They modify fundamental behaviors like quantity, duration, speed, and rotation of the particle cells.



The pEmitter controls

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Randomize and Random Seed

The Random Seed slider is used to seed all the variance and random number generators used by the node when creating the particle system. Two pEmitter nodes with exactly the same settings for all controls and the same random seed will generate exactly the same particle system. Changing the random seed will cause variation between the nodes. Click the Randomize button to automatically set a randomly chosen value for the Random Seed.

Number

This control is used to set the amount of new particles generated on each frame. A value of 1 would cause one new particle to be generated each frame. By frame 10, there would be a total of 10 particles in existence (unless Particle Lifespan was set to fewer than 10 frames).

Animate this parameter to specify the number of particles generated in total. For example, if only 25 particles in total are desired, animate the control to produce five particles on frame 0–4, then set a key on frame five to generate zero particles for the remainder of the project.

Number Variance

This modifies the amount of particles generated for each frame, as specified by the Number control. For example, if Number is set to 10.0 and Number Variance is set to 2.0, the emitter will produce anywhere from 9-11 particles per frame. If the value of Number Variance is more than twice as large as the value of Number, it is possible that no particles will be generated for a given frame.

Lifespan

This control determines how long a particle will exist before it disappears or 'dies.' The default value of this control is 100 frames, although this can be set to any value. The timing of many other particle controls is relative to the Lifespan of the particle. For example, the size of a particle can be set to increase over the last 80% of its life, using the Size Over Life graph in the Style tab of the pEmitter.

Lifespan Variance

Like Number Variance, the Lifespan Variance control allows the Lifespan of particles produced to be modified. If Particle Lifespan was set to 100 frames and the Lifespan Variance to 20 frames, particles generated by the emitter would have a lifespan of 90–110 frames.

Color

This provides the ability to specify from where the color of each particle is derived. The default setting is Use Style Color, which will provide the color from each particle according to the settings in the Style tab of the pEmitter node.

The alternate setting is Use Color From Region, which overrides the color settings from the Style tab and uses the color of the underlying bitmap region.

The Use Color From Region option only makes sense when the pEmitter region is set to use a bitmap produced by another node in the composition. Particles generated in a region other than a bitmap region will be rendered as white when the Use Color From Region option is selected.

Position Variance

This control determines whether or not particles can be 'born' outside the boundaries of the pEmitter region. By default, the value is set to zero, which will restrict the creation area for new particles to the exact boundaries of the defined region. Increasing this control's value above 0.0 will allow the particle to be born slightly outside the boundaries of that region. The higher the value, the 'softer' the region's edge will become.

Temporal Distribution

In general, an effect is processed per frame, based on the comp frame rate. However, processing some particles only at the exact frame boundaries can cause pulsing. To make the behavior subtly more realistic, the particles can be birthed in subframe increments.

The default, At The Same Time setting renders on frame boundaries, where as the other two settings take advantage of sub frame rendering. Randomly Distributed randomizes birth times +/- around the frame number, eg birth 10 particles at random sub times 24.1 24.85, 24.21, 24.37 etc. one particle at a time. Evenly Distributed births particles at regular sub times, eg 10 particles, birth 1 at at time at 24.0, 24.1, 24.2, 24.3, 24.4, 24.5 ... 24.8, 24.9.

These settings are influenced by the Sub Frame Accuracy setting in the pRender node. The Sub Frame Accuracy slider controls how many in-between frames are calculated between each frame. The higher the value the more accurate the particle calculation but the longer the render times.

Velocity

The controls in the Velocity section determine the speed and direction of the particle cells as the are generated from the emitter region.

Velocity and Velocity Variance

These determine the initial speed or velocity of new particles. By default, the particle has no velocity and will not move from its point of origin unless acted upon by outside forces. A velocity setting of 10.0 would cause the particle to cross the entire width of the image in one step so a velocity of 1.0 would cause the particle to cross the width of the image over 10 frames.

Velocity Variance modifies the velocity of each particle at birth, in the same manner described in Lifespan Variance and Number Variance above.

Inherit

Inherit Velocity passes the emitter region's velocity on to the particles. This slider has a wide range that includes negative and positive values. A negative value causes the particles to move in the opposite direction, a value of 1 will cause the particles to move with a velocity that matches the emitter region's velocity, and a value of 2 causes the particles to move ahead of the emitter region.

Angle and Angle Variance

This determines the angle at which particles with velocity applied will be heading at their birth.

Angle Z and Angle Z Variance

This is as above, except this control determines the angle of the particles along the Z space axis (toward or away from the camera).

Rotation

Rotation controls are used to set the orientation of particle cells and animating that orientation over time .

Rotation Mode

This menu control provides two options to help determine the orientation of the particles emitted. When the particles are spherical, the effect of this control will be unnoticeable.

- **Absolute Rotation:** The particles will be oriented as specified by the Rotation controls, regardless of velocity and heading.
- **Rotation Relative To Motion:** The particles will be oriented in the same direction as the particle is moving. The Rotation controls can now be used to rotate the particle's orientation away from its heading.

Rotation XYZ and Rotation XYZ Variance

These controls allow for Rotation of the individual particles. This can be particularly useful when dealing with a bitmap particle type, as the incoming bitmap may not be oriented in the desired direction.

Rotation XYZ Variance can be used to randomly vary the rotation by a specified amount around the center of the Rotation XYZ value to avoid having every particle oriented in the exact same direction.

Spin

Spin controls are auto animated controls that change the orientation of particle cells over time.

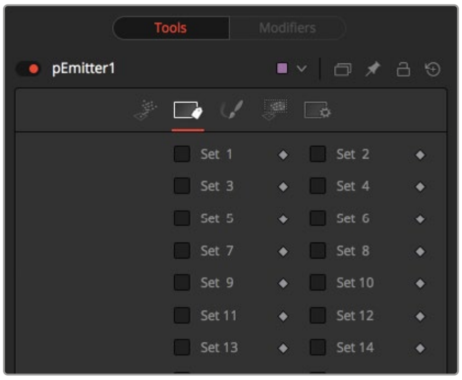
Spin XYZ and Spin Variance

These provide a spin to be applied to each particle at birth. The particles will rotate ‚x' degrees each frame, as determined by the value of Spin XYZ.

The Spin XYZ variances will vary the amount of rotation applied to each frame in the manner described by Number Variance and Lifespan Variance documented above.

Sets Tab

This tab contains settings that affect the physics of the particles emitted by the node. These settings do not directly affect the appearance of the particles. Instead, they modify behavior like velocity, spin, quantity, and lifespan.



The pEmitter Sets tab

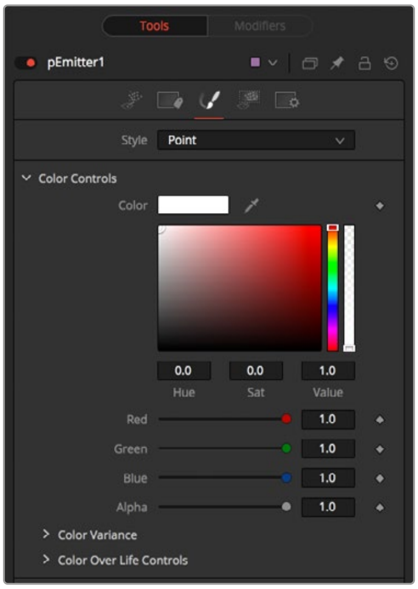
Set 1-32

To assign the particles created by a pEmitter to a given set, simply select the checkbox of the set number you want to assign. A single pEmitter node can be assigned to one or multiple sets. Once they are assigned in the pEmitter, you can enable sets in other particle nodes so they only affect particles from specific pEmitters.

Style Tab

The Style tab provides controls that affect the appearance of the particles. For detailed information about the style Tab, see the "The Common Controls" section at the end of this chapter.

FUSION



The pEmitter Style tab

Region Tab

The Region tab controls the shape, size, and location of the area that emits the particle cells. This is often called the Emitter. Only one emitter region can be set for a single pEmitter node. If the pRender is set to 2D, then the emitter region will produce particles along a flat plane in Z Space. 3D emitter regions possess depth and can produce particles inside a user-defined, three-dimensional region. For more detail on the Region tab, see “The Common Controls” section at the end of this chapter.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pFlock [pFl]

The pFlock node

pFlock Node Introduction

The pFlock node can be used to simulate the behavior of organic systems, such as a flock of birds or a colony of ants. Its use can make an otherwise mindless particle system appear to be motivated, or acting under the direction of intelligence.

The pFlock node works through two basic principles. Each particle attempts to stay close to other particles and each particle attempts to maintain a minimum distance from other particles.

The strength of these “desires” produces the seemingly motivated behavior perceived by the viewer.

FUSION

Inputs

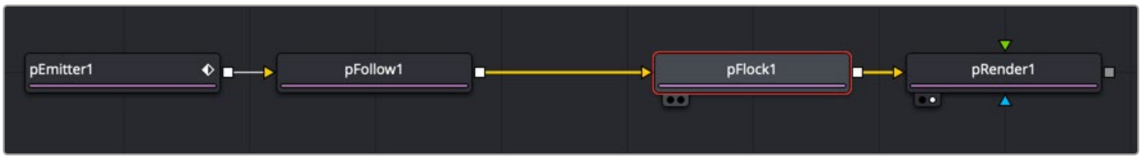
The pFlock node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green or magenta bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange background input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the flocking takes effect.

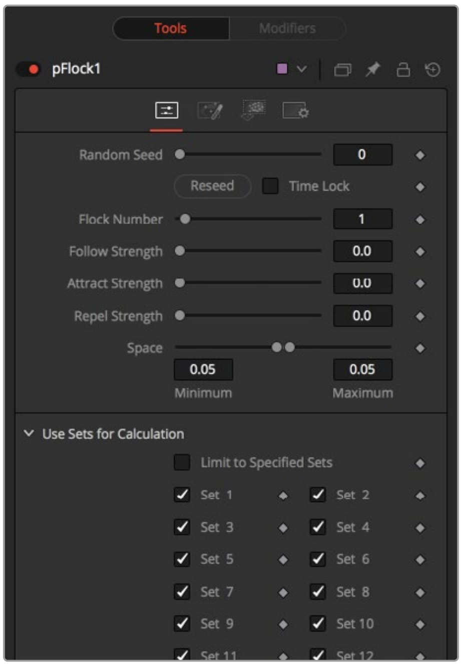
Basic Node Setup

When combined with pFollow, the pFlock node can produce natural swarming behaviors that change direction.



A pFlock node applying more herd-type mentality to particles

Inspector



The pFlock controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Flock Number

The value of this control represents the number of other particles that the affected particle will attempt to follow. The higher the value, the more visible “clumping” will appear in the particle system and the larger the groups of particles will appear.

Follow Strength

This value represents the strength of each particle’s desire to follow other particles. Higher values will cause the particle to appear to expend more energy and effort to follow other particles. Lower values increase the likelihood that a given particle will break away from the pack.

Attract Strength

This value represents the strength of attraction between particles. When a particle moves farther from other particles than the Maximum Space defined in the pFlock node, it will attempt to move closer to other particles. Higher values cause the particle to maintain its spacing energetically, resolving conflicts in spacing more rapidly.

Repel Strength

This value represents the force applied to particles that get closer together than the distance defined by the Minimum Space control of the pFlock node. Higher values will cause particles to move away from neighboring particles more rapidly, shooting away from the pack.

Minimum/Maximum Space

This range control represents the distance each particle attempts to maintain between it and other particles. Particles will attempt to get no closer or farther than the space defined by the Minimum/Maximum values of this range control. Smaller ranges will give the appearance of more organized motion. Larger ranges will be perceived as disorganized and chaotic.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pFollow [pFo]

The pFollow node

pFollow Node Introduction

Inserting the pFollow node into a particle branch causes the particles to spring back and forth toward a follow object. The follow object can be positioned in 3D or animated to create a new motion path for the particles.

Inputs

The pFollow node has a single orange input by default. Like most particle nodes, this orange background input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

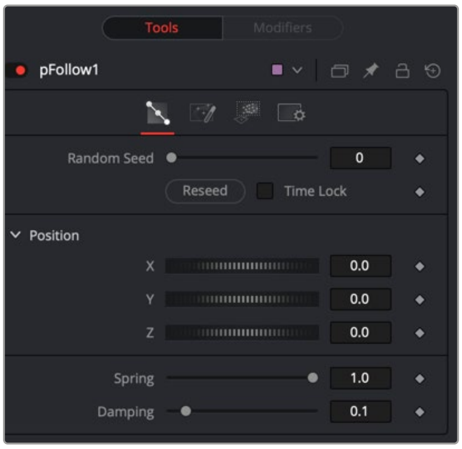
Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where particles will follow the position point.

Basic Node Setup

When combined with pFlock, the pFollow node can produce natural swarming behaviors that change direction.

A pFollow node introduces a follow object that influences the particles' motion.

Inspector



The pFollow Controls tab

Random Seed

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Position XYZ

The position controls are used to create the new path by positioning the follow object. Moving the XYZ parameters displays the onscreen position of the follow object. Animating these parameters creates the new path the particles will be influenced by.

Spring

The Spring setting causes the particles to move back and forth along the path. The spread of the spring motion increases over the life of the particles depending on the distance between the particles and the follow object. Higher spring settings increase the elasticity, while lower settings decrease elasticity.

Dampen

This value attenuates the spring action. A lower setting offers less resistance to the back and forth spring action. A higher setting applies more resistance.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pFriction [pFr]

The pFriction node

pFriction Node Introduction

The pFriction node applies resistance to the motion of a particle, slowing the particle's motion through a defined region. This node produces two types of Friction. One type reduces the velocity of any particle intersecting/crossing the defined region, and one reduces or eliminates spin and rotation.

Inputs

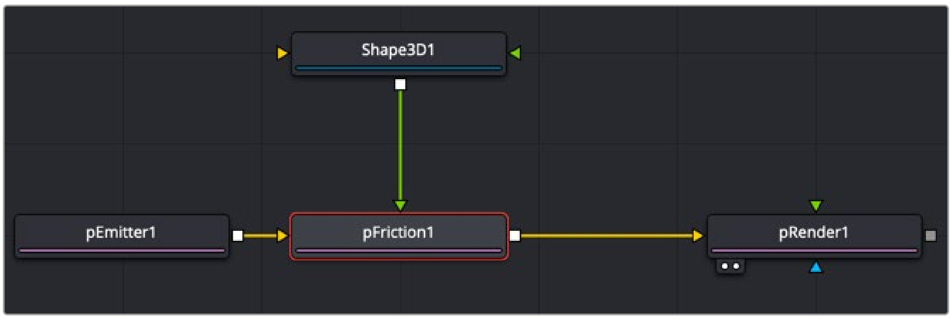
The pFriction node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green or magenta bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the friction occurs.

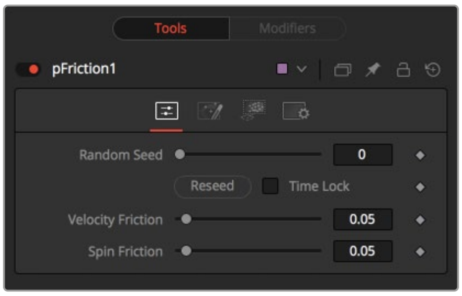
Basic Node Setup

The pFriction node is placed in between the pEmitter and pRender. A Shape 3D node is used to create the region the particles bounce off.



A pFriction node using a Shape 3D node as the region where friction is introduced to the particles.

Inspector



The pFriction controls

Random Seed

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Velocity Friction

This value represents the Friction force applied to the particle's Velocity. The larger the value, the greater the friction, thus slowing down the particle.

Spin Friction

This value represents the Friction force applied to the particle's Rotation or Spin. The larger the value, the greater the friction, thus slowing down the rotation of the particle.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pGradientForce [pGF]

The pGradient Force node

pGradient Force Node Introduction

The particles are affected by a force generated by the gradients in the alpha values of the input image. Particles will accelerate along the gradient, moving from white to black (high values to low values).

This node can be used to give particles the appearance of moving downhill or following the contour of a provided shape.

Inputs

The pGradient Force node accepts two inputs: the default orange input from a particle node and one from a bitmap image with an alpha channel gradient. A magenta or teal bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

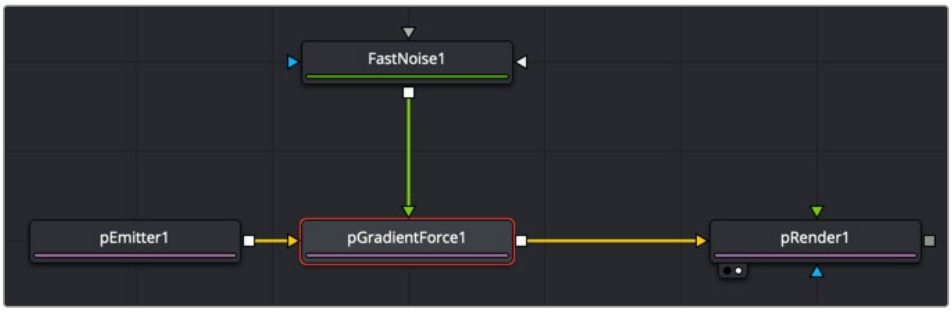
Input: The orange input takes the output of other particle nodes.

Input: The green input takes the 2D image that contains the alpha channel gradient.

Region: The magenta or teal region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the gradient force occurs.

Basic Node Setup

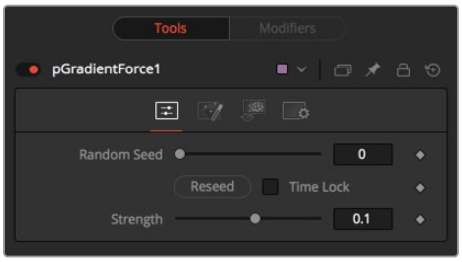
The pGradient Force node is placed in between the pEmitter and pRender nodes. A Fast Noise node is used to create the alpha gradient used to modify the velocity of the particles.



A pGradient Force node using a Fast Noise node as the gradient to modify the particles' motion

FUSION

Inspector



The pGradient Force controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result.

Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Strength

Gradient Force has only one specific control, which affects the strength of the force and acceleration applied to the particles. Negative values on this control will cause the Gradient Force to be applied from black to white (low values to high values).

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pImage Emitter [pIE]

The pImage Emitter node

pImage Emitter Node Introduction

The pImage Emitter node takes an input image and treats each pixel of the image as if it were a particle. The main differences between the pImage Emitter and the normal pEmitter is that instead of emitting particles randomly within a given region, this node emits pixels in a regular 2D grid with colors based on the input image.

Inputs

The pImage Emitter node has three inputs. Like most particle nodes, the orange input accepts only other particle nodes. Green and magenta inputs are 2D image inputs for custom image calculations. Optionally, there are teal or white bitmap or mesh inputs, which appear on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

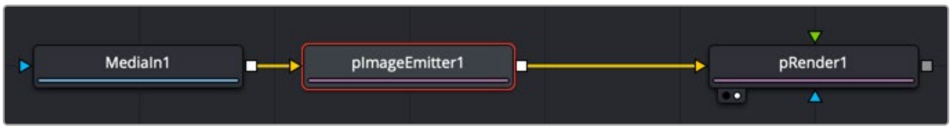
Input: Unlike most other particle nodes, the orange input on the pImage Emitter accepts a 2D image used as the emitter of the particles. If a region is defined for the emitter, this input is used to define the color of the particles.

Style Bitmap Input: This image input accepts a 2D image to use as the particles' image. Since this image duplicates into potentially thousands of particles, it is best to keep these images small and square—for instance, 256 x 256 pixels.

Region: The teal or white region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the particles are emitted.

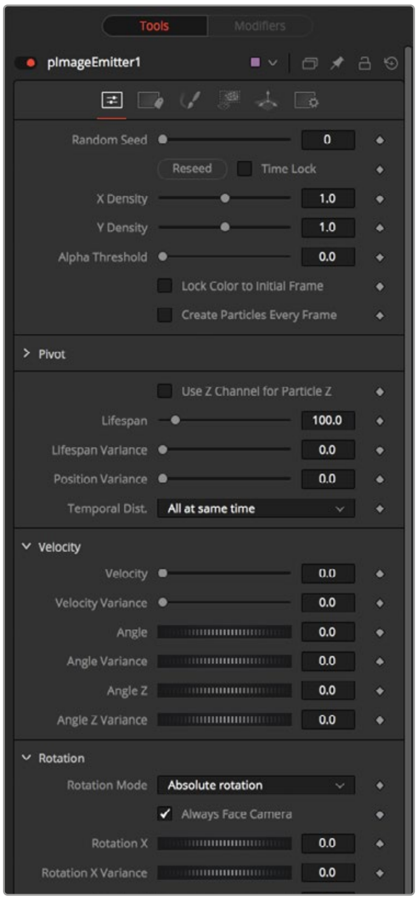
Basic Node Setup

The pImage Emitter node is placed at the start of a particle branch, replacing the location of a pEmitter node. Below, a MediaIn node is used to emit particles using the colors from the clip.



A pImage Emitter node emits particles based on an image connected to the orange input.

Inspector



The pImage Emitter controls

FUSION

The great majority of controls in this node are identical to those found in the pEmitter, and those controls are documented in that previous section. Below are the descriptions of the controls unique to the pImage Emitter node.

X and Y Density

The X and Y Density sliders are used to set the mapping of particles to pixels for each axis. They control the density of the sampling grid. A value of 1.0 for either slider indicates 1 sample per pixel. Smaller values will produce a looser, more pointillistic distribution of particles, while values above 1.0 will create multiple particles per pixel in the image.

Alpha Threshold

The Alpha Threshold is used for limiting particle generation so that pixels with semitransparent alpha values will not produce particles. This can be used to harden the edges of an otherwise soft alpha channel. The higher the threshold value, the more opaque a pixel must be before it will generate a particle. Note that the default threshold of 0.0 will create particles for every pixel, regardless of alpha, although many may be transparent and invisible.

Lock Particle Color to Initial Frame

Select this checkbox to force the particles to keep the color with which they were born throughout the life of the particle. If this is off, and the input image changes on successive frames, the particles will also change color to match the image. This allows video playback on a grid of particles.

Create Particles Every Frame

Enabling this creates a whole new set of particles every frame, instead of just one set on the frame. This can lead to very large particle systems but allows some interesting effects—for example, if the particles are given some initial velocity or if emitting from an animated source. Try a small velocity, an Angle Z of -90, and a seething Fast Noise as a source to get smoothly varying clouds of particles that you could fly through. Note that if this checkbox is left off, only one set of particles is ever created, and thus animating any of the emitter's other controls will have no effect.

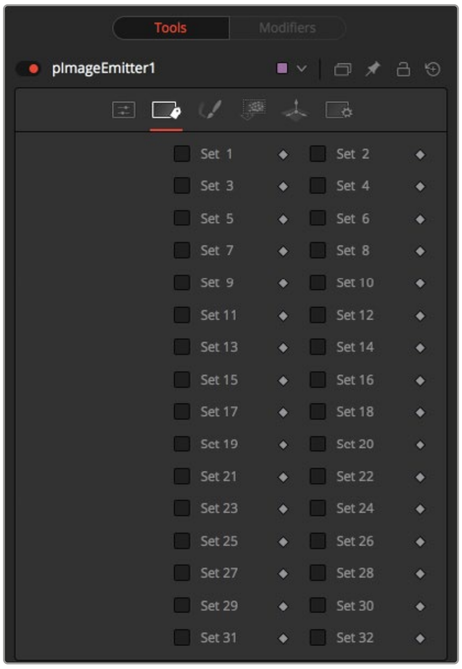
X/Y/Z Pivot

These controls allow you to position the grid of emitted particles.

Use Z Channel for Particle Z

If the input image used to generate the particles has a Z depth channel, that channel can be used to determine the initial position of the particle in Z space. This can have an interesting hollow shell effect when used in conjunction with camera rotation in the pRender node.

Sets Tab



The pImage Emitter Sets tab

NOTE: Pixels with a black (transparent) alpha channel will still generate invisible particles, unless you raise the Alpha Threshold above 0.0. This can slow down rendering significantly.

An Alpha Threshold value of 1/255 = 0.004 is good for eliminating all fully transparent pixels.

The pixels are emitted in a fixed-size 2D grid on the XY plane, centered on the Pivot position. Changing the Region from the default of All allows you to restrict particle creation to more limited areas. If you need to change the size of this grid, use a Transform 3D node after the pRender.

Remember that the various emitter controls apply only to particles when they are emitted. That is, they set the initial state of the particle and do not affect it for the rest of its lifespan. Since pImageEmitter (by default) emits particles only on the first frame, animating these controls will have no effect. However, if the Create Particles Every Frame checkbox is turned on, new particles will be emitted each frame and will use the specified initial settings for that frame.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

FUSION

pKill [pKl]

The pKill node

pKill Node Introduction

The Kill node is used to destroy (kill) any particle that crosses or intersects its region. It has no specific controls, as it has only one possible affect on a particle. The controls found in the Region tab are normally used to limit this node by restricting the effect to particles that fall within a certain region, age, set, or by reducing the probability of the node applying to a given particle.

Inputs

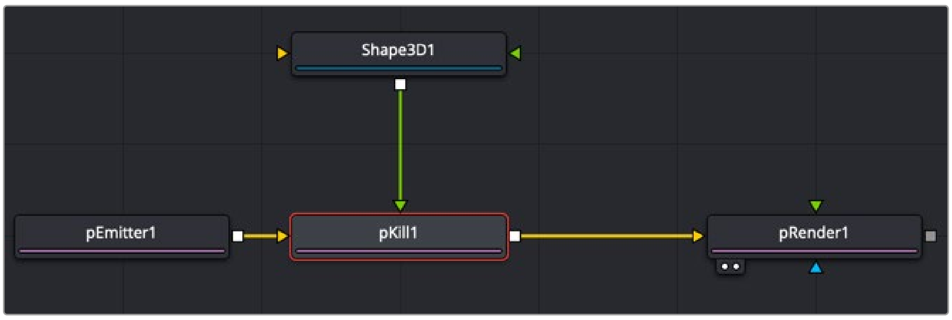
The pKill node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area particles are killed.

Basic Node Setup

The pKill node is placed in between the pEmitter and pRender nodes. A Shape 3D node is used to create the region where the particles die.



A pKill node using a Shape 3D node as the region where particles die

Inspector

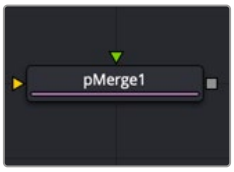
This node only contains common controls in the Conditions and Regions tabs. The Conditions and Regions controls are used to define the location, age, and set of particles that are killed.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pMerge [pMg]



The pMerge node

pMerge Node Introduction

This node has no controls whatsoever. It serves to combine particles from two streams. Any nodes downstream of the pMerge node will treat the two streams as one.

The combined particles will preserve any sets assigned to them when they were created, making it possible for nodes downstream of the pMerge to isolate specific particles when necessary.

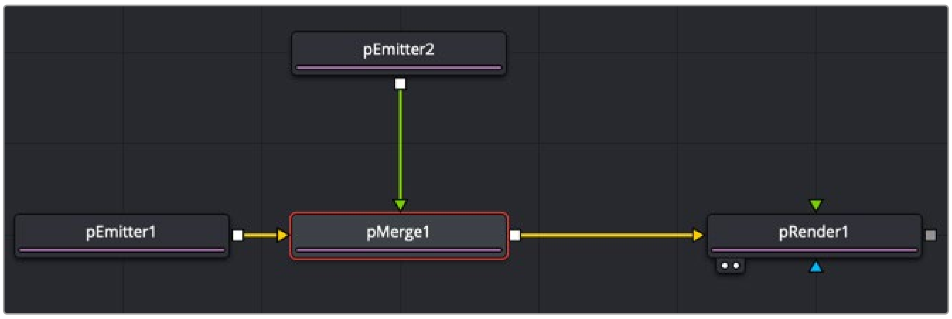
Inputs

The pMerge node has two identical inputs, one orange and one green. These two inputs accept only other particle nodes.

Particle 1 and 2 Input: The two inputs accept two streams of particles and merge them.

Basic Node Setup

The pMerge node connects two pEmitter nodes. The output of the pMerge can go on to feed other particle nodes or to a pRender.



A pMerge node combining two pEmitter nodes.

pPoint Force [pPF]

The pPoint Force node

pPoint Force Node Introduction

This node applies a force to the particles that emanate from a single point in 3D space. The pPoint Force can either attract or repel particles within its sphere of influence. There are four controls specific to the pPoint Force node.

Inputs

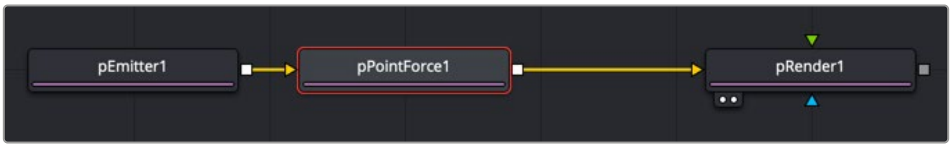
The pPoint Force node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the point force affects the particles.

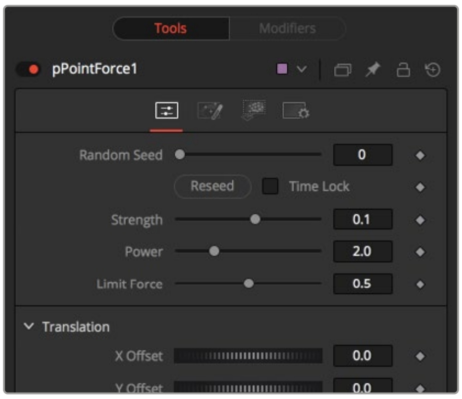
Basic Node Setup

The pPoint Force node is inserted between a pEmitter and a pRender node.



The pPoint Force node positions a tangent force that particles are attracted to or repelled from.

Inspector



The pPoint Force controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Strength

This parameter sets the Strength of the force emitted by the node. Positive values represent attractive forces; negative values represent repellent forces.

Power

This determines the degree to which the Strength of the force falls off over distance. A value of zero causes no falloff of strength. Higher values will impose an ever-sharper falloff in strength of the force with distance.

Limit Force

The Limit Force control is used to counterbalance potential problems with temporal sub-sampling. Because the position of a particle is sampled only once a frame (unless sub-sampling is increased in the pRender node), it is possible that a particle can overshoot the Point Force's position and end up getting thrown off in the opposite direction. Increasing the value of this control reduces the likelihood that this will happen.

X, Y, Z Center Position

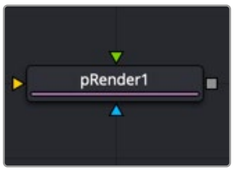
These controls are used to represent the X, Y, and Z coordinates of the point force in 3D space.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pRender [pRn]



The pRender node

pRender Node Introduction

The pRender node converts the particle system to either an image or geometry. The default is a 3D particle system, which must be connected to a Renderer 3D to produce an image. This allows the particles to be integrated with other elements in a 3D scene before they are rendered.

FUSION

Inputs

The pRender node has one orange input, a green camera input, and a blue effects mask input. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Camera Input: The optional green camera input accepts a camera node directly or a 3D scene with a camera connected that is used to frame the particles during rendering.

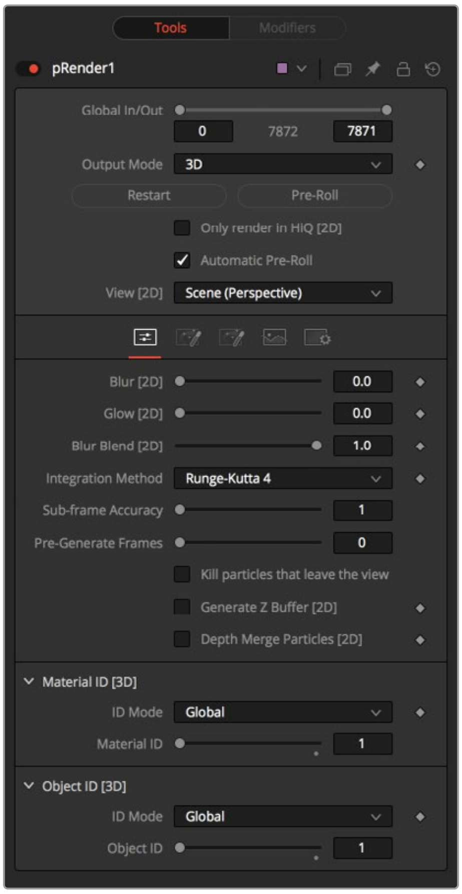
Effect Mask: The optional blue input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input for 2D particles crops the output of the particles so they are seen only within the mask.

Basic Node Setup

The pRender node is always placed at the end of a particle branch. If the pRender is set to 2D, then the output connects to other 2D nodes like a Merge node. If the pRender is set to 3D, the output connects to a 3D node like a Merge 3D.

All particle branches end with a pRender node.

Inspector



The pRender controls

FUSION

Output Mode (2D/3D)

While the pRender defaults to 3D output, it can be made to render a 2D image instead. This is done with the 3D and 2D buttons on the Output Mode control. If the pRender is not connected to a 3D-only or 2D-only node, you can also switch it by selecting View > 2D Viewer from the viewer's pop-up menu.

In 3D mode, the only controls in the pRender node that have any effect at all are Restart, Pre-Roll and Automatic Pre-Roll, Sub-Frame Calculation Accuracy, and Pre-Generate frames. The remaining controls affect 2D particle renders only. The pRender node also has a Camera input on the node tree that allows the connection of a Camera 3D node. This can be used in both 2D and 3D modes to allow control of the viewpoint used to render an output image.

Render and the Viewers

When the pRender node is selected in a node tree, all the onscreen controls from Particle nodes connected to it are presented in the viewers. This provides a fast, easy-to-modify overview of the forces applied to the particle system as a whole.

Pre-Roll Options

Particle nodes generally need to know the position of each particle on the last frame before they can calculate the effect of the forces applied to them on the current frame. This makes changing current time manually by anything but single frame intervals likely to produce an inaccurate image.

The controls here are used to help accommodate this by providing methods of calculating the intervening frames.

Restart

This control also works in 3D. Clicking on the Restart button will restart the particle system at the current frame, removing any particles created up to that point and starting the particle system from scratch at the current frame.

Pre-Roll

This control also works in 3D. Clicking on this button causes the particle system to recalculate, starting from the beginning of the render range up to the current frame. It does not render the image produced. It only calculates the position of each particle. This provides a relatively quick mechanism to ensure that the particles displayed in the views are correctly positioned.

If the pRender node is displayed when the Pre-Roll button is selected, the progress of the pre-roll is shown in the viewer, with each particle shown as point style only.

Automatic Pre-Roll

Selecting the Automatic Pre-Roll checkbox causes the particle system to automatically pre-roll the particles to the current frame whenever the current frame changes. This prevents the need to manually select the Pre-Roll button whenever advancing through time in jumps larger than a single frame. The progress of the particle system during an Automatic Pre-Roll is not displayed to the viewers to prevent distracting visual disruptions.

About Pre-Roll

Pre-Roll is necessary because the state of a particle system is entirely dependent on the last known position of the particles. If the current time were changed to a frame where the last frame particle state is unknown, the display of the particle is calculated on the last known position, producing inaccurate results.

To demonstrate:

1. Add a pEmitter and a pRender node to the composition.
2. View the pRender in one of the viewers.
3. Set the Velocity of the particles to 0.1.
4. Place the pEmitter on the left edge of the screen.
5. Set the Current Frame to 0.
6. Set a Render Range from 0–100 and press the Play button.
7. Observe how the particle system behaves.
8. Stop the playback and return the current time to frame 0.
9. In the pRender node, disable the Automatic Pre-Roll option.
10. Use the current time number field to jump to frame 10, and then to frames 60 and 90.

Notice how the particle system only adds to the particles it has already created and does not try to create the particles that would have been emitted in the intervening frames. Try selecting the Pre-Roll button in the pRender node. Now the particle system state is represented correctly.

For simple, fast-rendering particle systems, it is recommended to leave the Automatic Pre-Roll option on. For slower particle systems with long time ranges, it may be desirable to only Pre-Roll manually, as required.

- **Only Render in Hi-Q**
  Selecting this checkbox causes the style of the particles to be overridden when the Hi-Q checkbox is deselected, producing only fast rendering Point-style particles. This is useful when working with a large quantity of slow Image-based or Blob-style particles. To see the particles as they would appear in a final render, simply enable the Hi-Q checkbox.

- **View**
  This drop-down list provides options to determine the position of the camera view in a 3D particle system. The default option of Scene (Perspective) will render the particle system from the perspective of a virtual camera, the position of which can be modified using the controls in the Scene tab. The other options provide orthographic views of the front, top, and side of the particle system.

  It is important to realize that the position of the onscreen controls for Particle nodes is unaffected by this control. In 2D mode, the onscreen controls are always drawn as if the viewer were showing the front orthographic view. (3D mode gets the position of controls right at all times.)

  The View setting is ignored if a Camera 3D node is connected to the pRender node's Camera input on the node tree, or if the pRender is in 3D mode.

Conditions

- **Blur, Glow, and Blur Blend**
  When generating 2D particles, these sliders apply a Gaussian blur, glows, and blur blending to the image as it is rendered, which can be used to soften the particles and blend them. The result is no different than adding a Blur after the pRender node in the node tree.

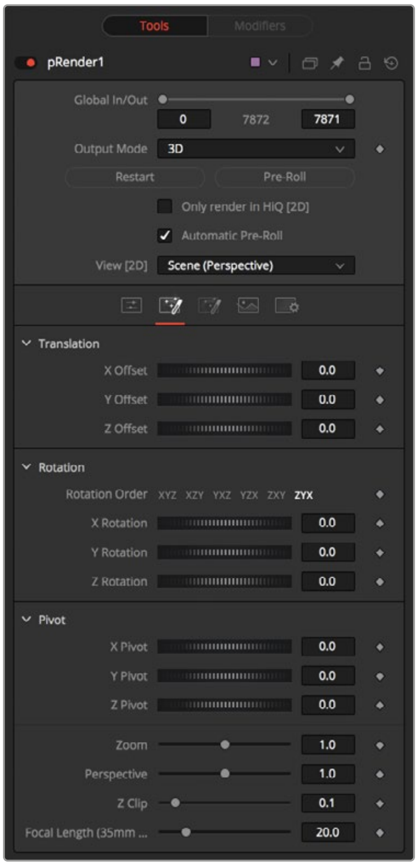
- **Sub-Frame Calculation Accuracy**
  This determines the number of sub-samples taken between frames when calculating the particle system. Higher values increase the accuracy of the calculation but also increase the amount of time to render the particle system.
- **Pre-Generate Frames**
  This control is used to cause the particle system to pre-generate a set number of frames before its first valid frame. This is used to give a particle system an initial state from which to start.

  A good example of when this might be useful is in a shot where particles are used to create the smoke rising from a chimney. Set Pre-Generate Frames to a number high enough to ensure that the smoke is already present in the scene before the render begins, rather than having it just starting to emerge from the emitter for the first few frames.
- **Kill Particles That Leave the View**
  Selecting this checkbox control automatically destroys any particles that leave the visible boundaries of the image. This can help to speed render times. Particles destroyed in this fashion never return, regardless of any external forces acting upon them.Generate Z Buffer

  Selecting this checkbox causes the pRender node to produce a Z Buffer channel in the image. The depth of each particle is represented in the Z Buffer. This channel can then be used for additional depth operations like Depth Blur, Depth Fog, and Downstream Z Merging.

  Enabling this option is likely to increase the render times for the particle system dramatically.
- **Depth Merge Particles**
  Enabling this option causes the particles to be merged using Depth Merge techniques, rather than layer-based techniques.

Scene Tab



The pRender Scene tab

FUSION

Z Clip

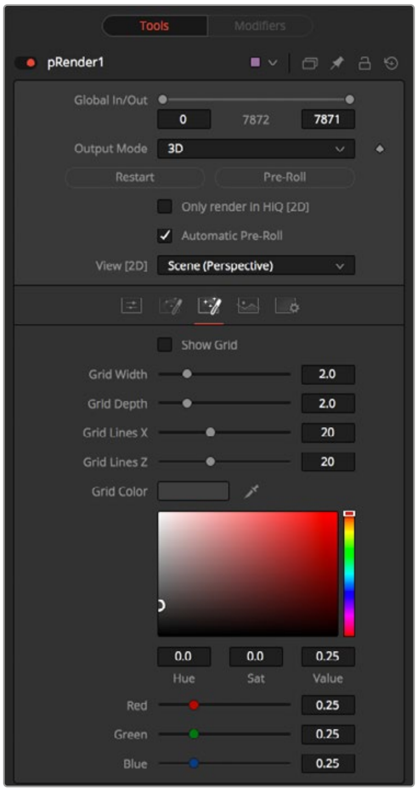
The Z Clip control is used to set a clipping plane in front of the camera. Particles that cross this plane are clipped, preventing them from impacting on the virtual lens of the camera and dominating the scene.

Grid Tab

These controls do not apply to 3D particles.

The grid is a helpful, non-rendering display guide used to orient the 2D particles in 3D space. The grid is never seen in renders, just like a center crosshair is never seen in a render. The width, depth, number of lines, and grid color can be set using the controls found in this tab.

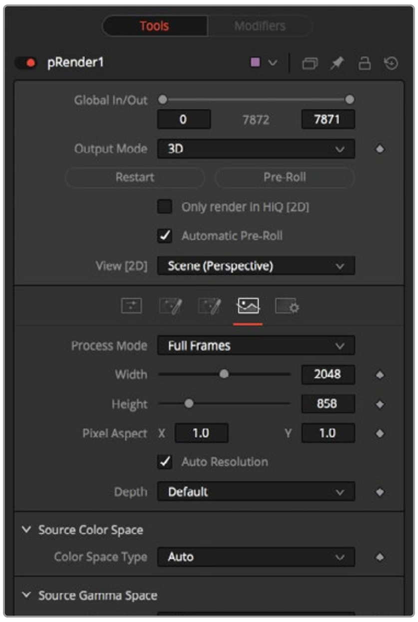
These controls cannot be animated.



The pRender Grid tab

Image Tab

The controls in this tab are used to set the resolution, color depth, and pixel aspect of the rendered image produced by the node.



The pRender Image tab

Process Mode

Use this menu control to select the Fields Processing mode used by Fusion to render changes to the image. The default option is determined by the Has Fields checkbox control in the Frame Format preferences.

Use Frame Format Settings

When this checkbox is selected, the width, height, and pixel aspect of the rendered images by the node will be locked to values defined in the composition's Frame Format preferences. If the Frame Format preferences change, the resolution of the image produced by the node will change to match. Disabling this option can be useful to build a composition at a different resolution than the eventual target resolution for the final render.

Width/Height

This pair of controls is used to set the Width and Height dimensions of the image to be rendered by the node.

Pixel Aspect

This control is used to specify the Pixel Aspect ratio of the rendered particles. An aspect ratio of 1:1 would generate a square pixel with the same dimensions on either side (like a computer display monitor), and an aspect of 0.9:1 would create a slightly rectangular pixel (like an NTSC monitor).

> **NOTE:** Right-click on the Width, Height, or Pixel Aspect controls to display a menu listing the file formats defined in the preferences Frame Format tab. Selecting any of the listed options will set the width, height, and pixel aspect to the values for that format, accordingly.

Depth

The Depth menu is used to set the pixel color depth of the particles. 32-bit pixels require 4X the memory of 8-bit pixels but have far greater color accuracy. Float pixels allow high dynamic range values outside the normal 0...1 range, for representing colors that are brighter than white or darker than black.

Source Color Space

You can use the Source Color Space menu to set the Color Space of the footage to help achieve a linear workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion, but rather adds the source space data into the metadata, if that metadata doesn't exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver, if explicit output spaces are defined there. There are two options to choose from:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Color Space Type menu where you can choose the correct color space of the image.

Source Gamma Space

Using the Curve type menu, you can set the Gamma Space of the footage and choose to remove it by way of the Remove Curve checkbox when working in a linear workflow. There are three choices in the Curve type menu:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Gamma Space Type menu where you can choose the correct gamma curve of the image.
- **Log:** Brings up the Log/Lin settings, similar to the Cineon tool.

For more information, *see Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 39 in the Fusion Reference Manual.

Remove Curve

Depending on the selected Gamma Space or on the Gamma Space found in Auto mode, the Gamma Curve is removed from, or a log-lin conversion is performed on, the material, effectively converting it to a linear output space.

Motion Blur

As with other 2D nodes in Fusion, Motion Blur is enabled from within the Settings tab. You may set Quality, Shutter Angle, Sample Center, and Bias, and Blur will be applied to all moving particles.

NOTE: Motion Blur on 3D mode particles (rendered with a Renderer 3D) also requires that identical motion blur settings are applied to the Renderer 3D node.

pSpawn [pSp]

The pSpawn node

pSpawn Node Introduction

The pSpawn node makes each affected particle act as an emitter that can produce one or more particles of its own. The original particle continues until the end of its lifespan, and each of the particles it emits becomes wholly independent with a lifespan and properties of its own.

As long as a particle falls under the effect of the pSpawn node, it will continue to generate particles. It is important to restrict the effect of the node with limiters like Start and End Age, Probability, Sets and Regions, and by animating the parameters of the emitter so that the node is operative only when required.

Inputs

By default, the pSpawn node has a single orange input. Like most particle nodes, this orange input accepts only other particle nodes. You can enable an image input by selecting Bitmap from the Style menu in the Style tab. Also, two region inputs, one for bitmap and one for mesh, appear on the node when you set the Region menu in the Region tab to either Bitmap or Mesh. The colors of these inputs change depending on the order they are enabled.

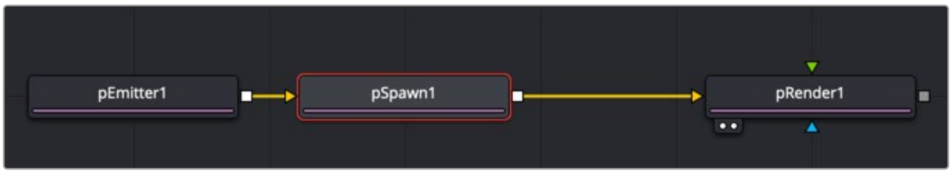
Input: The orange input accepts the output of other particle nodes.

Style Bitmap Input: This image input accepts a 2D image to use as the particles' image. Since this image duplicates into potentially thousands of particles, it is best to keep these images small and square—for instance, 256 x 256 pixels.

Region: The region inputs take a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the particles are emitted.

Basic Node Setup

The pSpawn node is placed between the pEmitter and pRender nodes. Using the Age parameter in the pSpawn's Conditions tab, you can spawn new particles as the old ones die off. This is one way to have a trail of particles shoot up in the air like a rocket and burst into sparkling fireworks.

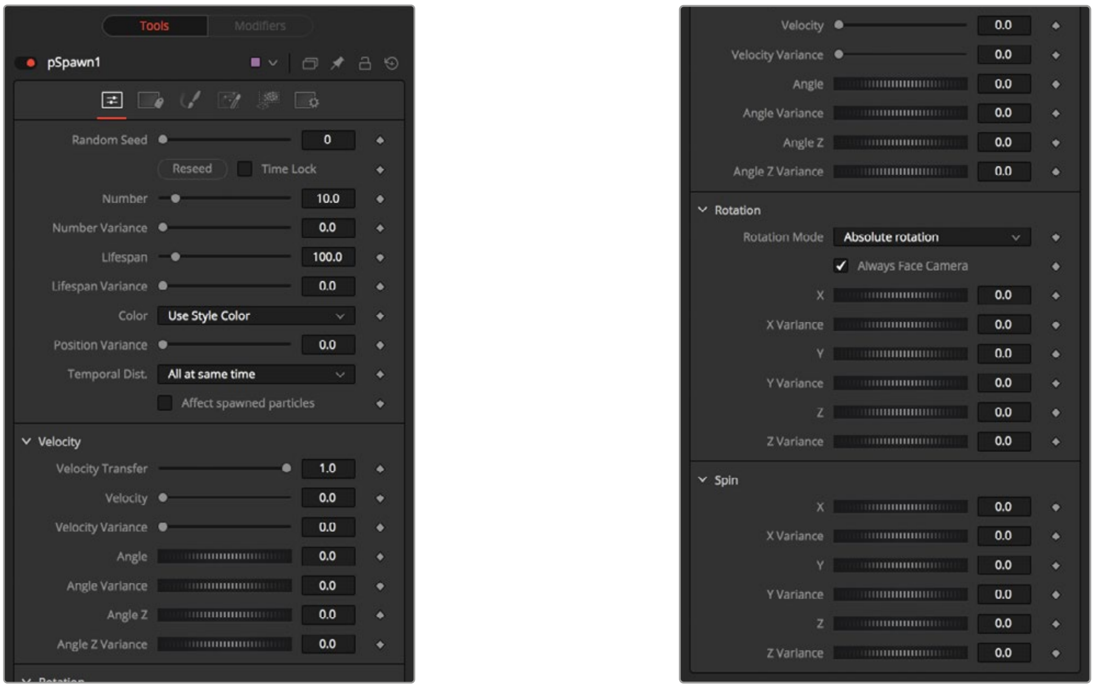


A pSpawn node used to generate new particles at specific points in the old particles' life

FUSION

Inspector

The pSpawn node has a large number of controls, most of which exactly duplicate those found within the pEmitter node. There are a few controls that are unique to the pSpawn node, and their effects are described below.



The pSpawn controls

FUSION

Affect Spawned Particles

Selecting this checkbox causes particles created by spawning to also become affected by the pSpawn node on subsequent frames. This can exponentially increase the number of particles in the system, driving render times up to an unreasonable degree. Use this checkbox cautiously.

Velocity Transfer

This control determines how much velocity of the source particle is transferred to the particles it spawns. The default value of 1.0 causes each new particle to adopt 100 percent of the velocity and direction from its source particle. Lower values will transfer less of the original motion to the new particle.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pTangent Force [pTF]

The pTangent Force node

pTangent Force Node Introduction

This node is used to apply a tangential force to the particles—a force that is applied perpendicularly to the vector between the pTangent Force's region and the particle it is affecting.

Inputs

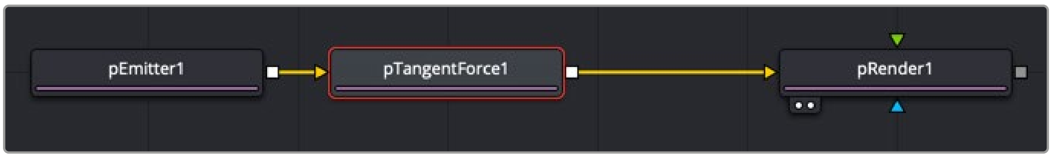
The pTangent Force node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area where the tangent force effects the particles.

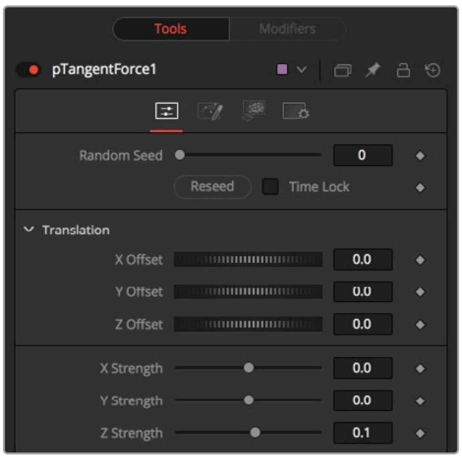
Basic Node Setup

The pTangent Force node is inserted between a pEmitter and a pRender node.



The pTangent Force node positions a tangent force that particles maneuver around.

Inspector



The pTangent Force controls

FUSION

The controls for this node are used to position the offset in 3D space and to determine the strength of the tangential force along each axis independently.

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result.

Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

X, Y, Z Center Position

These controls are used to represent the X, Y, and Z coordinates of the Tangent force in 3D space.

X, Y, Z Center Strength

These controls are used to determine the Strength of the Tangent force in 3D space.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in "The Common Controls" section at the end of this chapter.

pTurbulence [pTr]

The pTurbulence node

pTurbulence Node Introduction

The pTurbulence node imposes a frequency-based chaos on the position of each particle, causing the motion to become unpredictable and uneven. The controls for this node affect the strength and density of the Turbulence along each axis.

Inputs

The pTurbulence node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

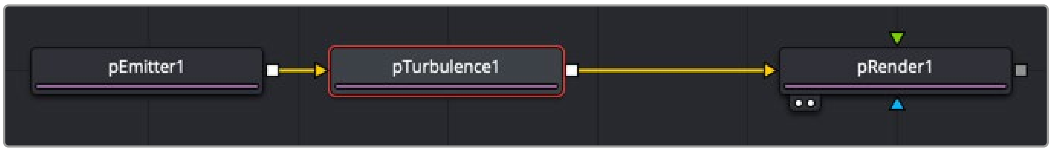
Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area of turbulence.

FUSION

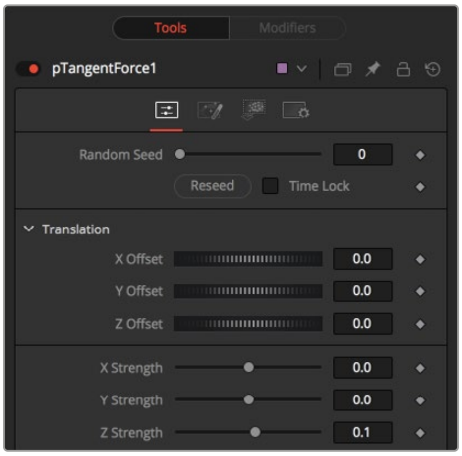
Basic Node Setup

The pTurbulence node is inserted between a pEmitter and a pRender node.



The pTurbulence node disturbs the rigid flow of particles for a more natural motion.

Inspector



The pTurbulence controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

X, Y, and Z Strength

The Strength control affects the amount of chaotic motion imparted to particles.

Strength Over Life

This mini Spline Editor control can be used to control the amount of turbulence applied to a particle according to its age. For example, a fire particle may originally have very little turbulence applied at the start of its life, and as it ages, the turbulence increases.

Density

Use this control to adjust the density in the turbulence field. Lower values causes more particle cells to be affected similarly, almost as if “waves” of the turbulence field run through the particles, affecting groups of cells at the same time. Higher values add finer variations to more individual particle cells causing more of a spread in the turbulence field.

Common Controls

Conditions, Style, Region, and Settings Tabs

The Conditions, Style, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in “The Common Controls” section at the end of this chapter.

pVortex [pVt]

The pVortex node

pVortex Node Introduction

The pVortex node applies a rotational force to each particle, causing them to be drawn toward the source of the Vortex. In addition to the Common Particle Controls, the pVortex node also has the following controls.

Inputs

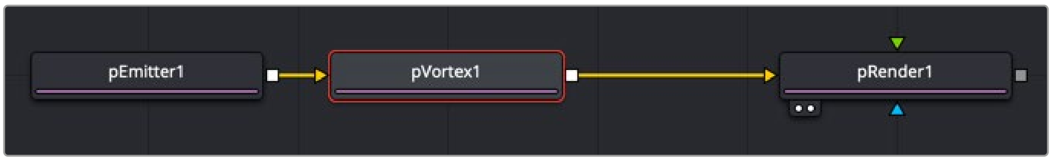
The pVortex node has a single orange input by default. Like most particle nodes, this orange input accepts only other particle nodes. A green bitmap or mesh input appears on the node when you set the Region menu in the Region tab to either Bitmap or Mesh.

Input: The orange input takes the output of other particle nodes.

Region: The green or magenta region input takes a 2D image or a 3D mesh depending on whether you set the Region menu to Bitmap or Mesh. The color of the input is determined by whichever is selected first in the menu. The 3D mesh or a selectable channel from the bitmap defines the area of the vortex.

Basic Node Setup

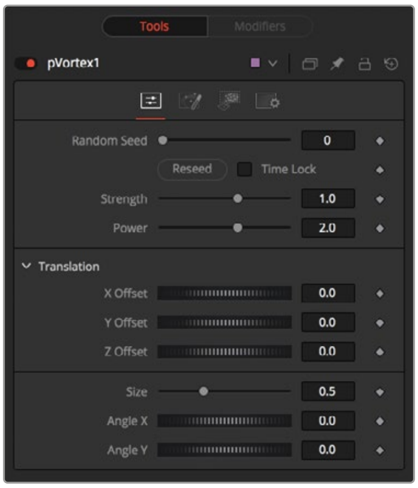
The pVortex node is placed in between the pEmitter and pRender nodes.



A pVortex node creates a spiraling motion for particles that fall within its pull.

FUSION

Inspector



The pVortex controls

Randomize

The Random Seed slider and Randomize button are presented whenever a Fusion node relies on a random result. Two nodes with the same seed values will produce the same random results. Click the Randomize button to randomly select a new seed value, or adjust the slider to manually select a new seed value.

Strength

This control determines the Strength of the Vortex Force applied to each particle.

Power

This control determines the degree to which the Strength of the Vortex Force falls off with distance.

X, Y, and Z Offset

Use these sliders to set the amount by which the vortex Offsets the affected particles.

Size

This is used to set the Size of the Vortex Force.

Angle X and Y

These sliders control the amount of rotational force applied by the Vortex along the X and Y axes.

Common Controls

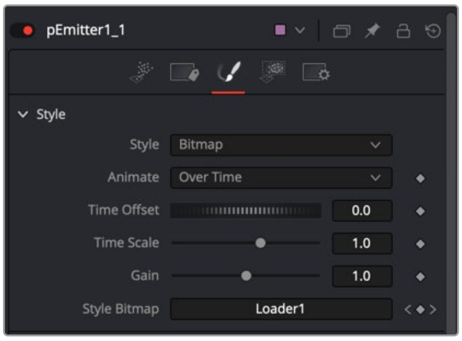
Conditions, Style, Region, and Settings Tabs

The Conditions, Region, and Settings tabs are common to all Particle nodes, so their descriptions can be found in the following “The Common Controls” section.

The Common Controls

Particle nodes share a number of identical controls in the inspector. This section describes the Style, Conditions, Region, and Settings tabs that are common among particle nodes.

Inspector



The pEmitter Style tab

Style Tab

The Style Tab is common to the pEmitter, pSpawn, pChangeStyle, and pImage Emitter. It controls the appearance of the particles using general controls like type, size, and color.

Style

The Style menu provides access to the various types of particles supported by the Particle Suite. Each style has its specific controls, as well as controls it will share with other styles.

- **Point:** This option produces particles precisely one pixel in size. Controls that are specific to Point Style are Apply Mode and Sub Pixel Rendered.

  Apply Mode: This menu applies only to 2D particles. 3D particle systems are not affected. It includes Apply modes for Add and Merge. Add combines overlapping particles by adding together the color values of each particle. Merge uses a simple over operation to combine overlapping particles.

  Sub Pixel Rendered: This checkbox determines whether the point particles are rendered with Sub Pixel precision, which provides smoother-looking motion but blurrier particles that take slightly longer to render.

- **Bitmap:** This style produces particle cells based on an image file or another node in the Node editor. When this option is selected an orange image input appears on the node in the node editor. There are several controls for affecting the appearance and animation. In addition to the controls in the Style section, a Merge section is displayed at the bottom of the inspector when Bitmap is selected as the Style. The Merge section includes controls for additive or subtractive merges when the particle cells overlap.

  Animate Over Time: This menu includes three options for determining how movie files play when they are used as particle cell bitmaps. The Over Time setting plays the movie file sequentially. For instance, when the comps is on frame 2, frame 2 of the movie file is displayed, when the comp is on frame 3, frame 3 of the movie files is displayed and so on. If a particle cell is not generated until frame 50, it begins with frame 50 of the movie file. This causes all particle cells to use the same image on any give frame of the comp. The Particle Age setting causes each particle cell to begin with the first frame of the movie file, regardless of when the particle cell is generated. The Particle Birth Time setting causes each particle to begin with the frame that coincides with the frame of the particle cell birth time. For instance, if the particle is generated on frame 25, then it uses frame 25 of the movie file for the entire comp. Unlike the other two options, the Particle Birth Time setting holds the same frame for the duration of the comp

Time Offset: This dial is used to slip or offset the starting frame used from the movie file. For instance, setting it to 10 will cause the movie file to start at frame 10 instead of frame 1.

Time Scale: This slider is a multiplier on the frame. Instead of using an offset, it changes the starting frame by multiplying the frame by the value selected with the slider. For instance, if a value of 2 is selected then when the playhead reaches frame 2, the movie files displays frame 4 (2x2=4) and when the playhead reaches frame 8, the movie file displays frame 16 (8x2=16).

Gain: The gain slider is a multiplier of the pixel value. It is used to apply a correction to the overall Gain of the Bitmap. Let's say you have a bitmap particle cell that contains a pixel value of R0.5 G0.5 B0.4 and you add a Gain of 1.2, you end up with a pixel value of R0.6 G0.6, B0.48 (i.e., 0.4 * 1.2 = 0.48) while leaving black pixels unaffected. Higher values produce a brighter image, whereas lower values reduce both the brightness and the transparency of the image.

Style Bitmap: This control appears when the Bitmap style is selected, along with an orange Style Bitmap input on the node's icon in the Node view. Connect a 2D node to this input to provide images to be used for the particles. You can do this on the Node view, or you may drag and drop the image source node onto the Style Bitmap control from the Node Editor or Timeline, or right-click on the control and select the desired source from the Connect To menu.

- **Blob:** This option produces large, soft spherical particles, with controls for Color, Size, Fade timing, Merge method, and Noise.

  Noise: This slider only applies to 2D Blob particles. The noise slider is used to introduce a computer generated Perlin noise pattern into the blob particles in order to give the blobs more texture. A setting of 0 introduces no noise to the Blob particles and a setting of 1 introduces the maximum amount of noise.

- **Brush:** This styles produces particle cells based on any image file located in the brushes directory. There are numerous controls for affecting the appearance and animation.

  Gain: The gain slider is a multiplier of the pixel value. It is used to apply a correction to the overall Gain of the image that is used as the Brush. Let's say you have a brush particle cell that contains a pixel value of R0.5 G0.5 B0.4 and you add a Gain of 1.2, you end up with a pixel value of R0.6 G0.6, B0.48 (i.e., 0.4 * 1.2 = 0.48) while leaving black pixels unaffected. Higher values produce a brighter image, whereas lower values reduce both the brightness and the transparency of the image.

  Brush: This menu shows the names of any image files stored in the Brushes directory. The location of the Brushes directory is defined in the Preferences dialog, under Path Maps. The default is the Brushes subdirectory within Fusion's install folder.

  Use Aspect From: The Use Aspect From menu includes three settings for the aspect ratio of the brush image. You can choose image format to use the brush image's native aspect ration. Choose Frame Format to use the aspect ratio set in the Frame Format Setting in the Fusion Preferences, or choose Custom to enter your own Pixel X and Y dimensions.

- **Line:** This style produces straight line-type particles with optional "falloff." The Size to Velocity control described below (under Size Controls) is often useful with this Line type. The Fade control adjusts the amount of falloff over the length of the line.

- **Point Cluster:** This style produces small clusters of single-pixel particles. Point Clusters are similar to the Point style; however, they are more efficient when a large quantity of particles is required. This style shares parameters with the Point style. Additional controls specific to Point Cluster style are Number of Points and Number Variance.

  Sub Pixel Rendered: This checkbox determines whether the point particles are rendered with Sub Pixel precision, which provides smoother-looking motion but blurrier particles that take slightly longer to render.

  Number of Points and Variance: The value of this control determines how many points are in each Point Cluster.

Color Controls

The Color Controls select the color and Alpha values of the particles generated by the emitter.

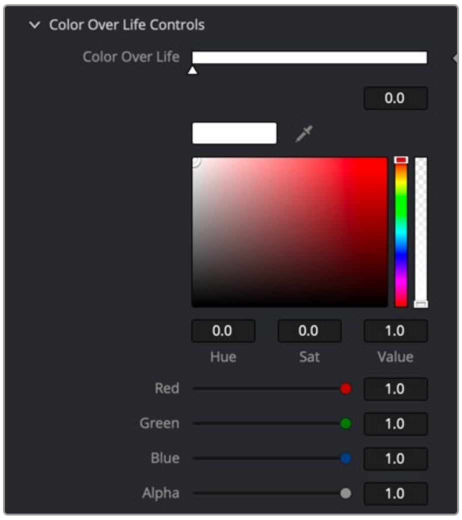
Color Variance

These range controls provide a means of expanding the colors produced by the pEmitter. Setting the Red variance range at -0.2 to +0.2 will produce colors that vary 20% on either side of the red channel, for a total variance of 40%. If the pEmitter is set to produce R0.5, G0.5, B0.5 (pure gray), the variance shown above will produce points with a color range between R0.3, G0.5, B0.5, and R0.7, G0.5, B0.5.

To visualize color space as values between 0-256 or as 0-65535, change the values used by Fusion using the Show Color As option provided in the General tab within the Preferences dialog.

Lock Color Variance

This checkbox locks the color variance of the particles. Unlocking this allows the color variance to be applied differently to each color channel, giving rise to a broader range of colors.



Particles Color Over Life controls

Color Over Life

This standard gradient control allows for the selection of a range of color values to which the particle will adhere over its lifetime.

The left point of the gradient represents the particle color at birth. The right point shows the color of the particle at the end of its lifespan.

Additional points can be added to the gradient control to cause the particle color to shift throughout its life.

This type of control can be useful for fire-type effects (for example, the flame may start blue, turn orange, and end a darker red). The gradient itself can be animated over time by right-clicking on the control and selecting Animate from the contextual menu. All points on the gradient will be controlled by a single Color Over Life spline, which controls the speed at which the gradient itself changes. You may also use the From Image modifier, which produces a gradient from the range of colors in an image along a line between two points.

Particles Size and Fade controls

Size Controls

The majority of the Size Controls are self-explanatory. The Size and Size Variance controls are used to determine the size and degree of size variation for each particle. It is worth noting that the Point style does not have size controls (each point is a single pixel in size, and there is no additional control).

When a Bitmap Particle style is used, a value of 1.0 indicates that each particle should be the same size as the input bitmap. A value of 2.0 will scale the particle up in size by 200%. For the best quality particles, always try to make the input bitmap as big, or bigger, than the largest particle produced by the system.

For the Point Cluster style, the size control adjusts the density of the cluster, or how close together each particle will get.

There are additional size controls that can be used to adjust further the size of particles based on velocity and depth.

Size to Velocity

This increases the size of each particle relative to the velocity or speed of the particle. The velocity of the particle is added to the size, scaled by the value of this control.

1.0 on this control, such as for a particle traveling at 0.1, will add another 0.1 to the size (velocity * size to velocity + size = new size). This is most useful for Line styles, but the control can be used to adjust the size of any style.

Size Z Scale

This control measures the degree to which the size of each particle changes according to its Z position. The effect is to exaggerate or reduce the impact of perspective. The default value is 1.0, which provides a relatively realistic perspective effect.

Objects on the focal plane (Z = 0.0) will be actual-sized. Objects farther along Z will become smaller. Objects closer along Z will get larger.

A value of 2.0 will exaggerate the effect dramatically, whereas a value of 0.0 will cancel the effects of perspective entirely.

Size Over Life

This spline control determines the size of a particle throughout its lifespan. The vertical scale represents a percentage of the value defined by the Size control, from 0 to 200%. The horizontal scale represents a percentage of the particle's lifespan (0 to 100%).

This graph supports all the features available to a standard spline editor. These features can be accessed by right-clicking on the graph. It is also possible to view and edit the graph spline in the larger Spline Editor.

Fade Controls

This simple range slider provides a mechanism for fading a particle at the start and end of its lifetime. Increasing the Fade In value will cause the particle to fade in at the start of its life. Decreasing the Fade Out value will cause the particle to fade out at the end of its life.

This control's values represent a percentage of the particle's overall life, therefore, setting the Fade In to 0.1 would cause the particle to fade in over the first 10% of its total lifespan. For example, a particle with a life of 100 frames would fade in from frame 0…10.

Merge Controls

This set of particle controls affects the way individual particles are merged together. The Subtractive/Additive slider works as documented in the standard Merge node. The Burn-In control will cause the particles to overexpose, or "blow out," when they are combined.

None of the Merge controls will have any effect on a 3D particle system.

Particles Blur controls

Blur Controls

This set of particle controls can be used to apply a Blur to the individual particles. Blurring can be applied globally, by age, or by Z depth position.

None of the Blur controls will have any effect on a 3D particle system.

Blur (2D) and Blur Variance (2D)

These controls apply blur to each particle. Unlike the Blur in the pRender node, this is applied to each particle independently before the particles are merged together. The Blur Variance slider modifies the amount of blur applied to each particle.

Blur Over Life

This spline graph controls the amount of blur that is applied to the particle over its life. The vertical scale represents a percentage of the value defined by the Blur control. The horizontal scale represents a percentage of the particle's lifespan.

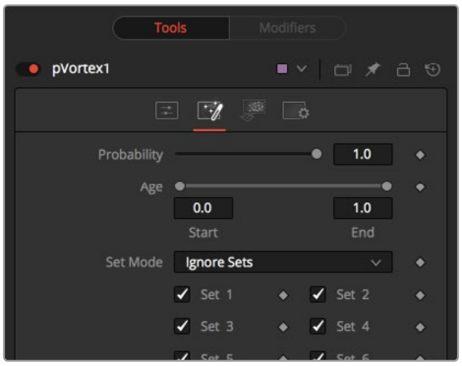
This graph supports all of the features available to a standard Spline Editor. These features can be accessed by right-clicking on the graph. It is also possible to view and edit the spline in the larger Spline editor.

Z Blur (DoF) (2D) and DoF Focus

This slider control applies blur to each particle based on its position along the Z axis.

The DoF Focus range control is used to determine what area of the image remains in focus. Lower values along Z are closer to the camera. Higher values are farther away. Particles within the range will remain in focus. Particles outside that range will have the blur defined by the Z Blur control applied to them.

Conditions



pVortex Conditions tab

Conditions Tab

The Conditions tab limits the particles that are affected by the node's behavior. You can limit the particle using probability or more specifically using sets.

Probability

The Probability slider determines the percentage of chance that the node affects any given particle.

The default value of 1.0 affects all particles. A setting of 0.6 would mean that each particle has a 60 percent chance of being affected by the control.

Probability is calculated for each particle on each frame. For example, a particle that is not affected by a force on one frame has the same chance of being affected on the next frame.

Start/End Age

This range control can be used to restrict the effect of the node to a specified percentage of the particle lifespan.

For example, to restrict the effect of a node to the last 20 percent of a particle's life, set the Start value to 0.8, and the End value remains at 1.0. The node on frames 80 through 100 only affects a particle with a lifespan of 100 frames.

Set Mode Menu

The Set Mode menu drives how the particle node influences the active particle sets. There are three options from this menu:

- **Ignore Sets:** The particle node disregards the state of the Set checkboxes and applies to all nodes.
- **Affect Specified Sets:** The particle node applies its behavior to the active Set checkboxes only.
- **Ignore Specified Sets:** The particle node applies its behavior to the inactive Set checkboxes only.

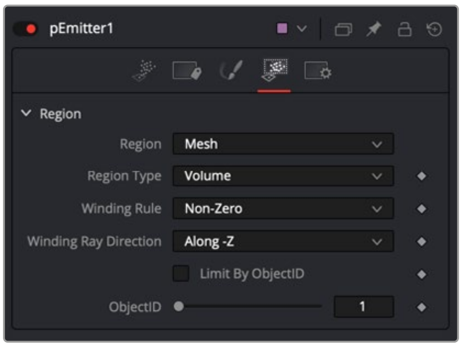
Set

The state of a Set # checkbox determines if the Particle node's effect will be applied to the particles in the set. It allows you to limit the effects of some nodes to a subset of particles.

Sets are assigned by the nodes that create particles. These include the pEmitter, pImage Emitter, pChangeStyle, and the pSpawn nodes.

Region Tab

The Region tab is used to restrict the node's effect to a geometric region or plane, and to determine the area where particles are created if it's a pEmitter node or where the behavior of a node has influence.



pEmitter Regions tab set to mesh

The Region tab is common to almost all particle nodes. In the pEmitter node Emitter Regions are used to determine the area where particles are created. In most other tools it is used to restrict the tool's effect to a geometric region or plane. There are seven types of regions, each with its own controls. Only one emitter region can be set for a single pEmitter node. If the pRender is set to 2D, then the emitter region will produce particles along a flat plane in Z Space. 3D emitter regions possess depth and can produce particles inside a user-defined, three-dimensional region.

Region Mode Menu

The Region Mode menu includes seven types of regions to define the area, each with its controls.

- **All:** In 2D, the particles will be created anywhere within the boundaries of the image. In 3D, this region describes a cube 1.0 x 1.0 x 1.0 units in size.

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- **Bézier:** Bézier mode uses a user-created polyline to determine the region where particles are created. The Bézier mode works in both 2D and 3D modes; however, the Bézier polyline region can only be created in 2D.

  To animate the shape of the polyline over time or to connect it to another polyline, right-click the Shape animation label at the bottom of the inspector and select the appropriate option from the drop-down menu.
- **Bitmap:** A Bitmap source from one of the other nodes in the composition will be used as the region where particles are born.
- **Cube:** A full 3D Cube is used to determine the region within which particles are created. The height, width, depth, and XYZ positions can all be determined by the user and be animated over time.
- **Line:** A simple line control determines where particles are created. The line is composed of two end-points, which can be connected to Paths or Trackers, as necessary. This type of emitter region includes X, Y, and Z position controls for the start and end of the line
- **Mesh:** Any 3D Mesh can be used as a region. In Mesh mode, the region can also be restricted by the Object ID using the ObjectID slider. See below for a more in-depth explanation of how mesh regions work.
- **Rectangle:** The Rectangle region type is like the Cube type, except that this region has no depth in Z space. Unlike other 2D emitter regions, this region can be positioned and rotated in Z space.
- **Sphere:** This is a spherical 3D emitter region with Size and Center Z controls. Sphere (3D) is the default region type for a new pEmitter node.

Mesh Regions

Region Type

The Region Type drop-down menu allows you to choose whether the region will include the inner volume or just the surface. For example, with a pEmitter mesh region, this determines if the particles emit from the surface or the full volume.

Winding Rule and Winding Ray Direction

The Winding Rule and Winding Ray Direction parameters determine how the mesh region handles particle creation with meshes that are not closed, as is common in many meshes imported from external applications. This scenario is common with imported mesh geometry, and even geometry that appears closed will frequently appear to “leak” thanks to improperly welded vertices.

To determine if a particle is in the interior of an object, a ray is cast from infinity through that particle and then out to -infinity. The Winding Ray Direction determines which direction this ray is cast in. Each time a surface is pierced by the ray, it is recorded and added onto a total to generate a winding number. Going against a surfaces normal counts as +1, and going with the normal counts as -1.

The Winding Rule is then used to determine what is inside/outside. For example, setting the Winding Rule to Odd means that only particles with odd values for the winding number are kept when creating the particles. The exact same approach is used to ensure that polylines that intersect themselves are closed properly.

For example, the following node tree and image show two image planes being used as a mesh region for particle creation.

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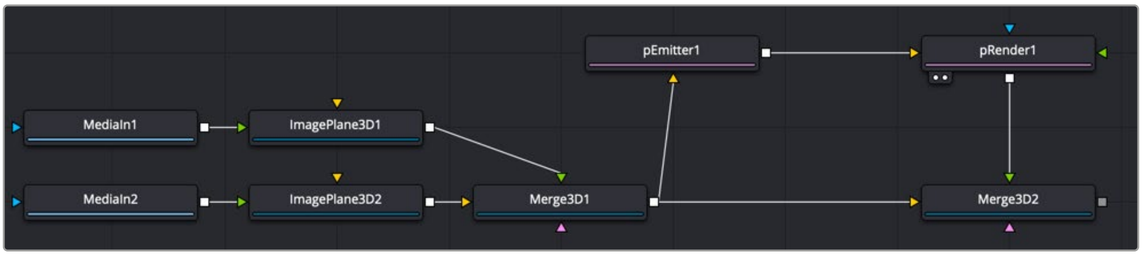
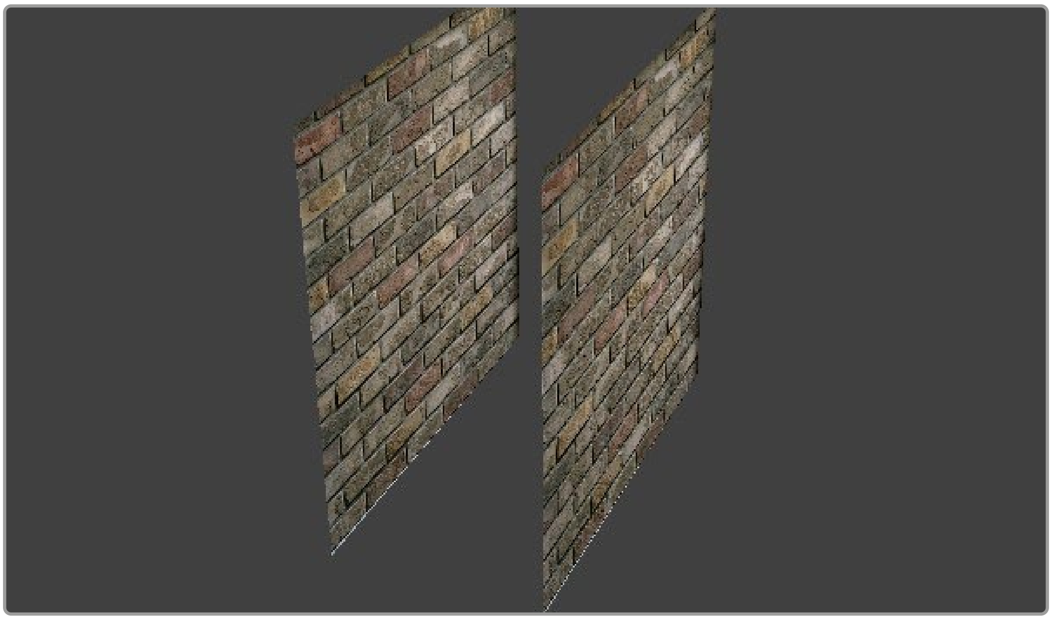
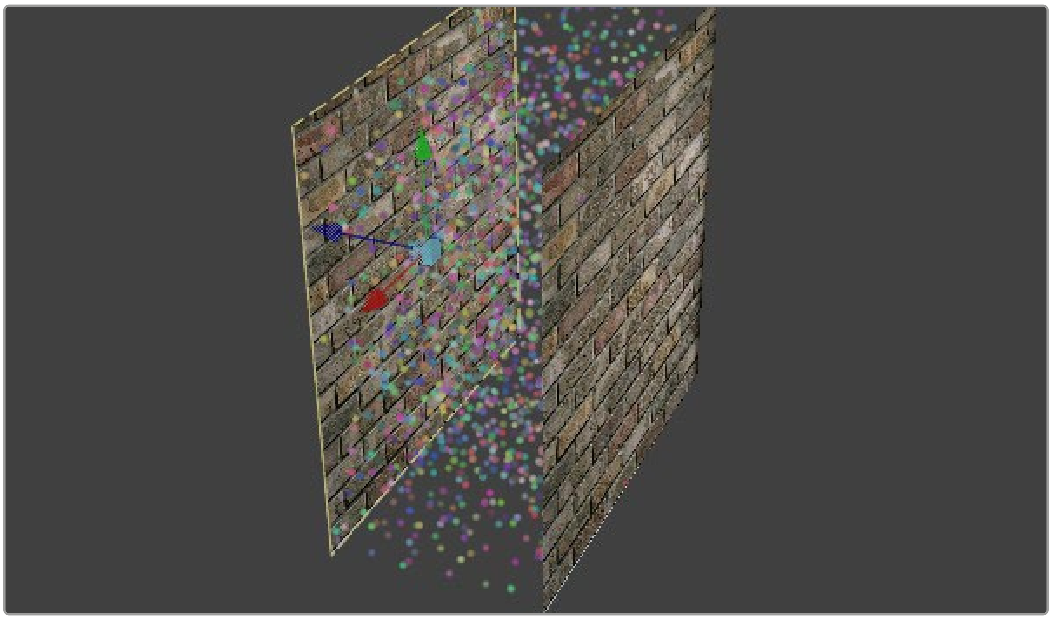




By setting the region's Winding Ray Direction to the Z (blue) axis, this mesh can then be treated as a closed volume for purposes of particle creation, as pictured below.



Limit By ObjectID

Selecting this checkbox allows the Object ID slider to select the ObjectID used as part of the region.

Style Tab

The Style tab exists in the pEmitter, pSpawn, pChangeStyle, and pImage Emitter. It controls the appearance of the particles, allowing the look of the particles to be designed and animated over time.

Style

The Style menu provides access to the various types of particles supported by the Particle Suite. Each style has its specific controls, as well as controls it will share with other styles.

- **Point Style:** This option produces particles precisely one pixel in size. Controls that are specific to Point style are Apply Mode and Sub Pixel Rendered.
- **Bitmap Style and Brush Style:** Both the Bitmap and Brush styles produce particles based on an image file. The Bitmap style relies on the image from another node in the node tree, and the Brush style uses image files in the Brushes directory. They both have numerous controls for affecting their appearance and animation, described below.
- **Blob Style:** This option produces large, soft spherical particles, with controls for Color, Size, Fade timing, Merge method, and Noise.
- **Line Style:** This style produces straight line-type particles with optional "falloff." The Size to Velocity control described below (under Size Controls) is often useful with this Line type. The Fade control adjusts the amount of falloff over the length of the line.
- **Point Cluster Style:** This style produces small clusters of single-pixel particles. Point Clusters are similar to the Point style; however, they are more efficient when a large quantity of particles is required. This style shares parameters with the Point style. Additional controls specific to Point Cluster style are Number of Points and Number Variance.

Style Options

The following options appear only on some of the styles, as indicated below.

Apply Mode (Point and Point Cluster)

This control applies only to 2D particles; 3D particle systems are not affected.

- **Add:** Overlapping particles are combined by adding together the color values of each particle.
- **Merge:** Overlapping particles are merged.

Sub Pixel Rendered (Point and Point Cluster)

This checkbox determines whether the point particles are rendered with Sub Pixel precision, which provides smoother-looking motion but blurrier particles that take slightly longer to render.

Number of Points and Variance (Point Cluster)

The value of this control determines how many points are in each Point Cluster.

Animate (Bitmap Style)

If the Bitmap source is a movie file or image sequence, this menu determines which frame is grabbed from the source and applied to newly-created particles.

- **Over Time:** All particles use the image produced by the Style Bitmap node at the current time, and change to each successive image together in step, as time increases. A particle created at frame 1 will contain the image at frame 1 of the Style Bitmap. At frame 2, the original particle will use the image from frame 2, and so will any new particles. All created particles will share the same bitmap image from their source at all times.
- **Particle Age:** Each particle animates through the sequence of images provided by the Style Bitmap node, independently of other particles. In other words, an individual particle's appearance is taken from the Style Bitmap node at successive times, indexed by its age.

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- **Particle Birth Time:** New particles take the image from the Style Bitmap node at the current time and keep it unchanged until the end of the particle's lifespan. Thus, particles generated on a given frame will all have the same appearance and will stay that way.

Time Offset (Bitmap Style)

This control allows the Bitmap source frame to be offset in time from the current frame.

Time Scale (Bitmap Style)

This control scales the time range of the source bitmap images by a specified amount. For example, a scale of 2 will cause the particle created at frame 1 to be read from the bitmap source at frame 2.

Gain (Bitmap and Brush Style)

This control applies a gain correction to the image used as the bitmap. Higher values produce a brighter image, whereas lower values reduce both the brightness and the transparency of the image.

Style Bitmap (Bitmap Style)

This control appears when the Bitmap style is selected, along with an orange Style Bitmap input on the node's icon in the Node view. Connect a 2D node to this input to provide images to be used for the particles. You can do this on the Node view, or you may drag and drop the image source node onto the Style Bitmap control from the Node Editor or Timeline, or right-click on the control and select the desired source from the Connect To menu.

Brush (Brush Style)

This menu shows the names of any image files stored in the Brushes directory. The location of the Brushes directory is defined in the Preferences dialog, under Path Maps. The default is the Brushes subdirectory within Fusion's install folder. If no images are in this directory, the only option in the menu will be None, and no particles will be rendered.

Noise (Blob Style)

Increasing this control's value will introduce grain-type noise to the blobby particle.

Fade (Line Style)

The Fade control adjusts the falloff over the line particle's length.

The default value of 1.0 causes the line to fade out completely by the end of the length.

Color Controls

The Color Controls select the color and Alpha values of the particles generated by the emitter.

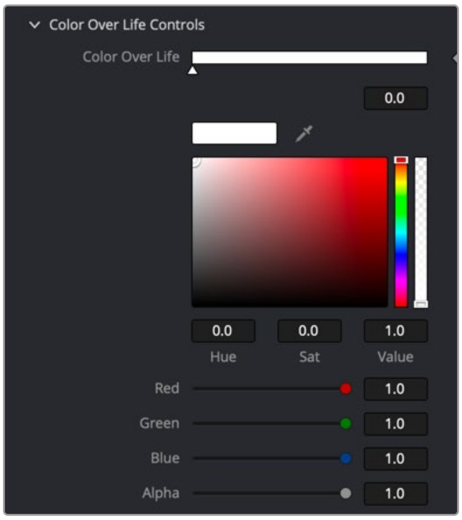
Color Variance

These range controls provide a means of expanding the colors produced by the pEmitter. Setting the Red variance range at -0.2 to +0.2 will produce colors that vary 20% on either side of the red channel, for a total variance of 40%. If the pEmitter is set to produce R0.5, G0.5, B0.5 (pure gray), the variance shown above will produce points with a color range between R0.3, G0.5, B0.5, and R0.7, G0.5, B0.5.

To visualize color space as values between 0-256 or as 0-65535, change the values used by Fusion using the Show Color As option provided in the General tab within the Preferences dialog.

Lock Color Variance

This checkbox locks the color variance of the particles. Unlocking this allows the color variance to be applied differently to each color channel, giving rise to a broader range of colors.



Particles Color Over Life controls

Color Over Life

This standard gradient control allows for the selection of a range of color values to which the particle will adhere over its lifetime.

The left point of the gradient represents the particle color at birth. The right point shows the color of the particle at the end of its lifespan.

Additional points can be added to the gradient control to cause the particle color to shift throughout its life.

This type of control can be useful for fire-type effects (for example, the flame may start blue, turn orange and end a darker red). The gradient itself can be animated over time by right-clicking on the control and selecting Animate from the contextual menu. All points on the gradient will be controlled by a single Color Over Life spline, which controls the speed at which the gradient itself changes. You may also use the From Image modifier, which produces a gradient from the range of colors in an image along a line between two points.